

EcoFed: Coverage-Constrained, Energy-Priced Federated Multi-Exit Learning for Battery-Limited Edge Fleets

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Abstract

Federated learning on wearable and IoT devices spends a scarce resource twice: battery energy when the model is trained, and again every time it is used. We study federated training of multi-exit networks, where a device can train only the first blocks to save energy, and ask how clients, depth, upload precision and local epochs should be chosen under per-device energy budgets. A convergence bound shows that depth heterogeneity creates an error floor that grows with the share of selected clients that skip a block, and gives the coverage needed for a target accuracy. We turn this into EcoFed, a drift-plus-penalty controller with per-device energy queues, a shadow energy price and block-coverage queues, which costs $O(N|\mathcal{C}|)$ per round. We evaluate it on three public benchmarks (UCI HAR, PAMAP2, Speech Commands; person-wise splits, 16 paired seeds, equal 14-trial tuning budgets for all eight methods) with an assumed, not measured, energy model. On label-skewed PAMAP2 at a binding budget EcoFed raises the final accuracy by 15 to 28 points over seven baselines and the early-stopped accuracy by 9 to 17 points over the energy-blind ones (Holm-corrected). On near-IID UCI HAR its early-stopped accuracy is within 1.1 points of all baselines while it uses up to 2.0 times less fleet energy, 1.7 to 4.2 times less energy to a target accuracy than FedAvg-type methods, and, at the two looser budgets, drains 0.4–9% instead of 16–31% of the devices for most baselines (at the tightest budget it drains more, 14%, because the baselines stop early). On Speech Commands it is worse than several baselines, and a static capability-based depth rule beats it at the tightest budget on PAMAP2 when training can be stopped at the best checkpoint. Ablations show that the energy queue and depth control are

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essential, that the coverage queue protects the deep exits, and the advantage grows when radio energy is expensive. Confidence-gated exits cut inference energy by 1.3 to 7.2 times, and serving overtakes training energy within 12 minutes to 1.7 days of fleet operation.

Keywords: Energy-aware AI, federated learning, multi-exit networks, adaptive inference, edge computing, wearable sensing

1. Introduction

Sensing devices that people wear or carry, such as watches, phones and fitness bands, generate continuous streams of inertial data that are useful for health monitoring, rehabilitation and context-aware services, and that are too personal and too voluminous to be collected centrally. Federated learning (FL) trains models across such devices without moving the raw data (McMahan et al., 2017; Kairouz et al., 2021). In practice, however, an FL round is not free for the device: local training draws on a battery that must also serve the user, and the model that comes out of training will run on the same device, around the clock, for as long as it is deployed. Energy is therefore the scarce resource of cross-device learning at the edge, and it enters twice—once when the model is trained and again, every second, when it is used.

The standard FL toolbox treats energy indirectly. Client-selection rules choose participants by statistical utility and speed (Lai et al., 2021); compression reduces the bits sent (Konečný et al., 2016; Alistarh et al., 2017); capacity-aware methods give weak devices a smaller sub-network (Diao et al., 2021; Kim et al., 2023; Ilhan et al., 2023); wireless-FL studies optimise transmit power and CPU frequency for a fixed model (Tran et al., 2019; Yang et al., 2021). Each acts on one knob, usually with a fixed assignment, and none of them asks what a Joule of training buys, how that answer differs between a fast, efficient device and a slow, inefficient one, or whether the cheaper updates still give a model that is trained everywhere it needs to be. On the deployment side, early-exit networks reduce inference energy by letting easy inputs leave early (Teerapittayanon et al., 2016; Kaya et al., 2019), but their training in federated, battery-limited fleets is rarely considered together with the energy it consumes.

The idea. A multi-exit network offers a natural handle on energy: a device that trains only the first d blocks pays roughly in proportion to d , and the

shallow blocks are the ones that every exit relies on. The risk is that the deep blocks, and the deep exits, are then trained too rarely. Our starting point is to treat this risk as a precisely quantifiable bias (Theorem 1): when selected clients train block j with probability σ_j , the aggregate update is biased by an amount that grows with $1 - \sigma_j$ and produces an error floor that no number of rounds removes. This turns a heuristic worry into a design constraint—keep σ_j above a target—and the constraint can be enforced online by a queue.

EcoFed. We build a controller, EcoFed, around that constraint. Each round, every client scores its feasible configurations (depth, upload precision, local epochs) by a utility, an energy cost priced by a per-client queue and a shadow price, and a bonus for reaching blocks whose coverage queue is backlogged; the server then picks the best few. Energy queues measure spending against a per-client budget in units of the budget itself, so weak, small-battery devices are priced out of expensive configurations without being excluded. Coverage queues keep deep blocks trained. The controller needs no model of the learning dynamics beyond a monotone utility and uses the energy that the device reports, so it keeps working when its own energy model is wrong (Section 6.6).

What we find. We evaluate on three public benchmarks with natural person-level partitions—two wearable inertial datasets and one keyword-spotting dataset—comparing against seven baselines (FedAvg, FedProx, quantised FedAvg, a paced FedAvg, static-depth heterogeneity, an Oort-style selector and an energy-greedy selector) under three battery tightnesses. All methods get the same tuning budget on separate seeds and validation subjects. The result is regime-dependent and we report it as such (Section 6): large accuracy gains on strongly label-skewed data with a binding budget, equal accuracy for much less energy on near-IID data, and no advantage—in places a disadvantage—on the keyword-spotting benchmark. We also analyse the lifecycle of the trained model, including the crossover point at which serving the model costs more energy than training it did, which shows how much room inference-side savings leave for training-side choices.

Scope and honesty about energy. All energy figures in this paper are *model-based*: constants for three device classes are taken from typical published orders of magnitude, and we did not have a power meter. We therefore treat robustness to energy-model error as a first-class experiment, and we state the consequence plainly: the paper establishes how a controller should behave

given an energy model and that it degrades gracefully when the model is wrong, not the absolute number of Joules on any particular board.

Contributions.

1. A device-aware energy model for federated multi-exit training and inference that separates arithmetic, memory traffic and radio energy, together with a lifecycle accounting that includes a break-even and a training–inference crossover (Section 3).
2. A convergence bound for federated training with heterogeneous client depth in which skipped exits create a coverage-dependent error floor, and a corollary that gives the coverage needed for a target accuracy (Section 4.1, Appendix A).
3. EcoFed, an online drift-plus-penalty controller that jointly chooses clients, depth, precision and epochs under per-client energy queues, a shadow energy price and coverage queues, with $O(N|\mathcal{C}|)$ cost per round (Section 4.2).
4. An evaluation on three benchmarks with equal tuning budgets and paired statistics, ablations, a test of the coverage prediction, robustness to energy-model error and to the radio-to-compute cost ratio, an energy–accuracy frontier, and a lifecycle analysis (Sections 5–6).

The remainder of the paper is organised as follows. Section 2 reviews related work, Section 3 defines the model, Section 4 presents theory and algorithm, Section 5 the experimental setup, Section 6 the results, Section 7 discusses implications and limitations.

2. Related work

Energy and carbon of machine learning. The energy cost of training large models and its carbon consequences have been documented for natural-language models (Strubell et al., 2019; Patterson et al., 2021) and, more broadly, across the lifecycle of AI systems (Wu et al., 2022). Reporting practices and tooling have followed (Henderson et al., 2020; You et al., 2023). Federated learning is not exempt: a first carbon-footprint study of federated training found that, depending on the configuration, it can emit up to two orders of magnitude more carbon than centralised training, and that in some settings it is comparable because embedded devices use little energy (Qiu

et al., 2023). Our work targets the edge end of this spectrum, where the binding constraint is not a data-centre power bill but the battery of each device, and where the energy of *later inference* is a first-order part of the lifecycle.

Efficient models for edge devices. TinyML research reduces the cost of inference through compact architectures, quantisation and pruning (Lin et al., 2020), and benchmark suites such as MLPerf Tiny treat energy per inference as a primary metric next to accuracy and latency (Banbury et al., 2021). Early-exit networks go one step further by making the cost of an inference depend on the input: easy samples leave at a shallow classifier and only hard samples reach the full network (Teerapittayanon et al., 2016; Kaya et al., 2019; Scardapane et al., 2020). Wearable activity recognition is a canonical application, because the signal is a continuous low-rate stream in which most windows are easy (Hammerla et al., 2016; Ordóñez and Roggen, 2016). These works optimise a model that is trained once in a data centre. We instead ask how a *multi-exit* model should be trained across a fleet of battery-limited devices, and what that training costs in energy relative to the inference savings it buys.

Federated learning under resource constraints. Federated averaging (McMahan et al., 2017) and its variants for heterogeneous data (Li et al., 2020a; Karimireddy et al., 2020; Reddi et al., 2021) are the baseline; convergence has been studied under non-IID data and partial participation (Li et al., 2020b). Communication has been attacked with quantisation and sparsification (Konečný et al., 2016; Alistarh et al., 2017). Resource-aware client selection chooses participants by statistical utility and speed (Lai et al., 2021), by loss (Cho et al., 2020), or with system-level constraints in mind (Bonawitz et al., 2019). A line of work models the energy of federated learning over wireless networks and optimises resource allocation, transmit power and CPU frequency (Tran et al., 2019; Yang et al., 2021). These studies fix the model each client trains; they choose *who* participates, *how long* or *how fast*. EcoFed also chooses *how much of the model* a client trains, together with precision and epochs, from a single energy price.

Heterogeneous-capacity federated learning. HeteroFL, DepthFL and ScaleFL let clients with different capabilities train sub-networks of different width or depth and aggregate them on the server (Diao et al., 2021; Kim et al., 2023; Ilhan et al., 2023). The sub-network of a client is typically *fixed* by a

Table 1: Positioning relative to representative prior work. ✓: addressed; ◦: partially; −: not addressed.

	Energy/battery model	Selection control	Depth control	Convergence with depth bias	Inference lifecycle
FedAvg/FedProx (McMahan et al., 2017; Li et al., 2020a)	−	−	−	−	−
Oort (Lai et al., 2021)	−	✓	−	−	−
Wireless-FL energy (Tran et al., 2019; Yang et al., 2021)	✓	◦	−	−	−
HeteroFL/DepthFL/ScaleFL (Diao et al., 2021; Kim et al., 2023; Ilhan et al., 2023)	−	−	◦ (static)	−	−
Early-exit networks (Teerapittayanon et al., 2016; Kaya et al., 2019)	−	−	−	−	✓
EcoFed (this paper)	✓	✓	✓ (dynamic)	✓	✓

capability class and there is no notion of a battery that is spent over time. Two gaps follow. First, when a client’s capacity changes with its remaining energy the assignment must be dynamic and must respect a long-term budget. Second, the effect of depth heterogeneity on convergence is usually argued empirically; we give a bound in which skipped exits create a bias floor and show how it can be controlled (Section 4.1). The static-depth baseline in our experiments follows the capability-class assignment used by this family.

Positioning. Table 1 summarises the differences. To our knowledge no previous work combines (i) a battery-constrained fleet with long-term energy queues, (ii) joint dynamic control of selection, depth, precision and epochs, (iii) a convergence bound that exposes the coverage floor and turns it into a constraint, and (iv) a lifecycle accounting that includes the inference energy of the trained multi-exit model. We state this as a claim about the works we are aware of rather than a certainty, since the literature on energy-aware federated learning is large and moving fast.

3. System model and problem formulation

This section fixes the learning model, the energy model, the battery dynamics and the lifecycle accounting used throughout the paper. Table 2 lists the notation.

3.1. Federated multi-exit learning

The global model is a multi-exit network with L blocks. Exit $j \in \{1, \dots, L\}$ is a classifier attached after block j , so its output depends on blocks $1, \dots, j$ only. Client k owns a local data distribution $\mathcal{D}_k$ and the loss of exit j is $f_k^j(\mathbf{w}) = \mathbb{E}_{(x,y) \sim \mathcal{D}_k} \ell(h_j(\phi_{1:j}(x; \mathbf{w})), y)$. The global objective is

$$F(\mathbf{w}) = \sum_{k=1}^N p_k \sum_{j=1}^L \lambda_j f_k^j(\mathbf{w}), \quad \lambda_j > 0, \quad \sum_j \lambda_j = 1. \quad (1)$$

Table 2: Notation.

Symbol	Meaning
N, T, m	clients, communication rounds, clients selected per round
L	number of blocks (and exits) of the multi-exit network
$\mathbf{w} = (\mathbf{w}_1, \dots, \mathbf{w}_L)$	parameters; block l includes the head of exit l
$c = (d, b, \tau)$	configuration: depth, upload bit-width, local epochs
f_k^j	loss of client k at exit j ; λ_j exit weights
$p_k = n_k / \sum_i n_i$	data share of client k
$e_k(c)$	energy of one participation of client k with configuration c (J)
$\bar{E}_k, B_k^t$	per-round energy budget and battery state of client k (J)
ϕ	budget tightness; fleet capacity $\mathcal{C} = \sum_k \bar{E}_k T$
q_k^t, Z_j^t	energy queue of client k ; coverage queue of exit j
$V, \lambda, \rho^{\min}$	utility weight, shadow energy price, target coverage
σ_j	probability that a selected client trains block j (coverage)

A participating client chooses a configuration $c = (d, b, \tau)$. It downloads blocks $1, \dots, d$, runs τ local epochs on the truncated objective $F_k^{(d)} = \sum_{j \leq d} \lambda_j f_k^j$, quantises its update to b bits with an unbiased stochastic quantiser and uploads the update of blocks $1, \dots, d$. Because $\partial f_k^j / \partial \mathbf{w}_l = 0$ for $l > j$, the gradient of $F_k^{(d)}$ with respect to block l is $\sum_{j=l}^d \lambda_j \nabla_l f_k^j$ and vanishes when $d < l$. The server aggregates every block separately over the clients that trained it (Section 4). A client of depth d therefore contributes to the shallow blocks fully, but it *withholds* the gradient contribution of exits $j > d$ to every block. This is the source of the coverage bias analysed in Section 4.1.

3.2. Energy model

The energy of one participation of client k on device class $u(k)$ is

$$e_k(c) = \underbrace{\tau n_k [\alpha_u 3 \text{MAC}_{1:d} + \beta_u \text{Mem}_{1:d}]}_{\text{local training}} + \underbrace{\varepsilon_u^{\text{tx}} b P_{1:d} + \varepsilon_u^{\text{rx}} 32 P_{1:d}}_{\text{communication}}, \quad (2)$$

where $\text{MAC}_{1:d}$ is the number of multiply-accumulates of one forward pass through blocks $1, \dots, d$ (the factor 3 accounts for the backward pass), $\text{Mem}_{1:d}$ is the number of bytes moved per sample (activations and weights, forward and backward), $P_{1:d}$ is the number of parameters in blocks $1, \dots, d$, α_u is the energy per MAC, β_u the energy per byte of memory traffic and $\varepsilon_u^{\text{tx}}, \varepsilon_u^{\text{rx}}$ the radio energy per transmitted and received bit. The term $\beta_u \text{Mem}$ separates

Table 3: Assumed device classes (fleet mix 40/40/20%). Energy per MAC α , per byte β , radio energy per bit, and relative battery size s_u .

Class	α (nJ/MAC)	β (nJ/B)	ε^{tx} (nJ/bit)	ε^{rx} (nJ/bit)	s_u
weak (Pi-class, Wi-Fi)	2.0	0.5	50	25	0.5
mid (Nano-class, Wi-Fi)	0.7	0.3	50	25	1.0
strong (Orin-class, LTE)	0.2	0.15	200	100	2.0

data movement from arithmetic, which matters because many edge workloads are memory-bound. The cost of one inference that exits at block j is $e_u^{\text{inf}}(j) = \alpha_u \text{MAC}_{1:j} + \beta_u \text{Mem}_{1:j}^{\text{inf}}$.

Device classes and an important caveat. We use three device classes whose constants follow the orders of magnitude of published figures for Raspberry-Pi-class, Jetson-Nano-class and Jetson-Orin-class boards (Table 3). *These constants are modelling assumptions, not measurements*: no power meter was available for this study, so every energy number in this paper is model-based. We therefore (i) report results over a sweep of radio-energy scales, (ii) stress the controllers with multiplicative errors between the constants assumed by the controller and the constants that actually drain the battery (Section 6.6), and (iii) check that the relative cost of depth predicted by the model matches the relative wall-clock cost on a real CPU (Section 6.9). Replacing Table 3 by constants fitted on hardware requires no change to any algorithm in this paper.

3.3. Batteries and budgets

Let $\bar{e} = \frac{m}{N} e^{\text{mid}}(L, 32, \tau_{\max}; \bar{n})$ be the average per-round energy of a fleet that selects m of N clients and runs the full configuration on a mid-class device. Client k receives a per-round budget $\bar{E}_k = \phi \bar{e} s_{u(k)}$ and an initial battery $B_k^0 = \bar{E}_k T$, so the *fleet capacity* is $\mathcal{C} = \sum_k \bar{E}_k T$. The parameter ϕ is the budget tightness: $\phi = 1$ would let every client run the full configuration at the nominal selection rate for the whole horizon, whereas $\phi = 0.1$ leaves roughly one tenth of that energy. A participation is feasible only if $B_k^t \geq e_k(c)$, and

$$B_k^{t+1} = B_k^t - e_k(c_k^t) \mathbb{I}[k \in \mathcal{S}_t], \quad B_k^t \geq 0. \quad (3)$$

In addition each client has a long-term budget $\limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t < T} \mathbb{E}[e_k(c_k^t) \mathbb{I}[k \in \mathcal{S}_t]] \leq \bar{E}_k$.

3.4. Lifecycle energy

After training, the model is deployed with a confidence-gated cascade: a sample exits at the first exit j whose maximum softmax probability is at least θ (the last exit is forced). With $\pi_j(\theta)$ the probability of exiting at j the expected inference energy is $\bar{e}^{\text{inf}}(\theta) = \sum_j \pi_j(\theta) e^{\text{inf}}(j)$, and the lifecycle energy of serving M inferences is

$$E_{\text{life}}(M) = E_{\text{train}} + M \bar{e}^{\text{inf}}(\theta). \quad (4)$$

For two models A and B with $\bar{e}_B^{\text{inf}} > \bar{e}_A^{\text{inf}}$ the *break-even number of inferences* after which A is cheaper over its lifecycle is

$$M^\star = \frac{E_{\text{train}}^A - E_{\text{train}}^B}{\bar{e}_{\text{inf},B} - \bar{e}_{\text{inf},A}}, \quad (5)$$

which is non-positive (A is cheaper from the first inference) whenever A also trains at lower cost. The *training-inference crossover* $M_\times = E_{\text{train}}/\bar{e}^{\text{inf}}$ is the number of inferences after which serving costs more energy than training did.

3.5. Problem statement

Given the fleet, the horizon T and the budgets, choose in every round the selected set $\mathcal{S}_t$ ($|\mathcal{S}_t| \leq m$) and the configurations c_k^t to maximise the accuracy of the final model for the energy actually consumed,

$$\max_{\{\mathcal{S}_t, c_k^t\}} \text{Acc}(\mathbf{w}^T) \quad \text{s.t. (3), } \frac{1}{T} \sum_t \mathbb{E}[e_k^t] \leq \bar{E}_k \quad \forall k, \quad \sigma_j \geq \rho_j^{\min} \quad \forall j. \quad (6)$$

The last constraint requires that every block keeps being trained with at least a target probability; Section 4.1 shows why it is necessary. Problem (6) is a mixed-integer programme coupled over time through the batteries and through the learning dynamics, so we solve it online with a Lyapunov controller. Besides the final accuracy we report accuracy at fixed fractions of the fleet capacity and the energy needed to reach a target accuracy, because a method that stops early must not be penalised for the energy it did not spend.

4. Coverage-constrained federated multi-exit learning

4.1. Why depth control needs a coverage constraint

Letting weak clients train only the first blocks saves energy, but it changes what the server optimises. If deep blocks are rarely trained the aggregated update is a *biased* estimate of ∇F . We quantify this bias and show that it produces an error floor that does not vanish with more rounds, and that the floor is controlled by the probability σ_j that a selected client trains block j .

Setting. Round t selects a random set $\mathcal{S}_t$ and a random depth d_k^t for each selected client, with a distribution that does not depend on the current iterate. Let $\sigma_{k,j} = \Pr[d_k^t \geq j \mid k \in \mathcal{S}_t]$ and $\sigma_j = \min_k \sigma_{k,j}$. Each selected client runs τ local SGD steps with step size η on $F_k^{(d_k)}$ and uploads the quantised model difference of its blocks. To isolate the effect of depth, the analysed aggregator weights the contribution of client k to block l by $p_k/\pi_{k,l}$ with $\pi_{k,l} = \Pr[k \in \mathcal{S}_t, d_k^t \geq l]$ (a Horvitz–Thompson weighting, so that the aggregate is unbiased within the covered set). The implementation uses the usual sample-size-normalised block averages; Section 6.4 tests the consequence of the theory for that practical aggregator.

Assumptions. (A1) Every f_k^j is L_s -smooth, hence F is L_s -smooth and bounded below by F^* . (A2) Stochastic gradients are unbiased and bounded, $\|\tilde{\nabla}\| \leq G$ almost surely, and $\|\nabla_l f_k^j(\mathbf{w})\| \leq G_l$ for all $k, j, \mathbf{w}$. (A3) The stochastic quantiser is unbiased with $\mathbb{E}\|Q(x) - x\|^2 \leq \omega_b \|x\|^2$, $\omega_b = O(2^{-2b})$. (A4) Conditionally on $\mathbf{w}^t$, the aggregated direction $\hat{g}_t$ has variance at most σ_h^2 (this collects minibatch noise, quantisation noise and client-sampling variance; it is finite whenever $\pi_{k,l} > 0$).

Theorem 1 (Convergence with heterogeneous depth). *Let $\gamma = \eta\tau$ and $\Delta = F(\mathbf{w}^0) - F^*$. Define the coverage bias*

$$B = \left(\sum_{l=1}^L B_l^2 \right)^{1/2}, \quad B_l = G_l \sum_{j>l} \lambda_j (1 - \sigma_j). \quad (7)$$

With $\tilde{\sigma}^2 = \sigma_h^2 + G^2/4$ and $\gamma = \min\{1/(4L_s), \sqrt{\Delta/(L_s \tilde{\sigma}^2 T)}\}$,

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla F(\mathbf{w}^t)\|^2 \leq \underbrace{6 \sqrt{\frac{L_s \tilde{\sigma}^2 \Delta}{T}} + \frac{16 L_s \Delta}{T}}_{\text{vanishes with } T} + \underbrace{6 B^2}_{\text{coverage floor}}. \quad (8)$$

The first group of terms is the usual non-convex rate. The second is a floor: it is zero if and only if $\sigma_j = 1$ for every $j \geq 2$ (every selected client trains every block) and it grows quadratically in the missing coverage $1 - \sigma_j$, weighted by the exit weights of the exits that are being skipped. The proof is in Appendix A.

Corollary 1 (Required coverage). *For L exits with equal weights $\lambda_j = 1/L$, $G_l \equiv G$ and $\sigma_j \equiv \sigma$ for $j \geq 2$, the floor equals $6B^2 = \frac{(L-1)(2L-1)}{L} G^2(1-\sigma)^2$. Hence a gradient-norm target ϵ is reachable only if*

$$\sigma \geq 1 - \sqrt{\frac{\epsilon L}{2(L-1)(2L-1)G^2}}. \quad (9)$$

Remark 1 (Interpretation). *Without a coverage guarantee a depth-heterogeneous method converges to a stationary point of the truncated objective it actually optimises, which need not be stationary for F . The theorem makes this precise and turns it into a design constraint: any energy-saving rule that shortens client depth must also guarantee that each block is trained with probability at least $\rho_j^{\min}$, and Eq. (9) tells how large $\rho^{\min}$ must be for a given accuracy target. This is why EcoFed carries a coverage queue (Section 4.2).*

4.2. The EcoFed controller

EcoFed solves (6) online with a drift-plus-penalty rule (Neely, 2010). Each client k keeps an *energy queue* q_k measured in units of its per-round budget, and the server keeps one *coverage queue* Z_j per exit:

$$q_k^{t+1} = \max\{q_k^t + e_k^t \mathbb{I}[k \in \mathcal{S}_t] / \bar{E}_k - 1, 0\}, \quad (10)$$

$$Z_j^{t+1} = \max\{Z_j^t + \rho_j^{\min} - \hat{\rho}_j^t, 0\}, \quad \hat{\rho}_j^t = \frac{\sum_{k \in \mathcal{S}_t} n_k \mathbb{I}[d_k^t \geq j]}{\sum_{k \in \mathcal{S}_t} n_k}. \quad (11)$$

Stability of q_k implies that the time-average energy of client k is at most $\bar{E}_k$; stability of Z_j implies that the realised coverage of block j is at least $\rho_j^{\min}$. The energy used in (10) is the energy *reported by the device*, so the queues self-correct when the assumed device constants are wrong.

Utility. Let $u_k = n_k \sqrt{\frac{1}{n_k} \sum_i \ell_i^2}$ be the statistical utility of client k (Lai et al., 2021), estimated from its last reported training loss (an optimistic constant for clients not yet seen), and $\tilde{u}_k = u_k / \bar{u}$ its value relative to the fleet mean. The utility of running configuration $c = (d, b, \tau)$ is

$$U_k(c) = \tilde{u}_k \left(\frac{d}{L}\right)^{\kappa_d} \left(1 - \frac{1}{2^b - 1}\right) \sqrt{\frac{\tau}{\tau_{\max}}}, \quad (12)$$

which increases with depth, precision and epochs with diminishing returns; $\kappa_d \in \{1, \frac{1}{2}\}$ is a tuned exponent ($\kappa_d < 1$ values shallow updates more).

Algorithm 1 EcoFed round t

Require: model $\mathbf{w}^t$, queues q_k, Z_j , parameters $V, \lambda, \rho^{\min}, m$

- 1: **for** each client k **do** ▷ runs on the server from cheap per-client reports
 - 2: $c_k^* \leftarrow \arg \max_{c: B_k \geq e_k(c)} s_k(c)$ by (13)
 - 3: **end for**
 - 4: $\mathcal{S}_t \leftarrow$ the $\leq m$ clients with the largest positive $s_k(c_k^*)$
 - 5: **for** each $k \in \mathcal{S}_t$ **in parallel do**
 - 6: train blocks $1..d_k$ for τ_k epochs, quantise to b_k bits, upload, report measured energy e_k and loss
 - 7: **end for**
 - 8: aggregate every block over the clients with $d_k \geq l$ (sample-size weighted); update B_k, q_k by (10), Z_j by (11)
-

Per-round decision. With a shadow price $\lambda \geq 0$ on energy (a constant added to every client's queue) each client scores its feasible configurations by

$$s_k(c) = V U_k(c) - (q_k + \lambda) \frac{e_k(c)}{\bar{E}_k} + \frac{1}{m} \sum_{j=1}^d Z_j, \quad c_k^* = \arg \max_{c: B_k \geq e_k(c)} s_k(c), \quad (13)$$

and the server selects the (at most) m clients with the largest strictly positive $s_k(c_k^*)$. A client whose best score is non-positive abstains, which is how the controller spends less than m clients' worth of energy in a round. The three terms are the utility reward, the energy cost priced by the queue plus the shadow price, and a coverage bonus: a configuration that reaches deep blocks earns $\frac{1}{m} \sum_{j \leq d} Z_j$, which is large exactly when those blocks have been under-trained. Because the coverage reward is additive across selected clients, the surrogate maximisation over $\mathcal{S}$ and $\{c_k\}$ is solved *exactly* by letting every client pick its best configuration and sorting; the per-round cost is $O(N|\mathcal{C}| + N \log N)$ with $|\mathcal{C}| = |\{1..L\}| \cdot |\{8, 32\}| \cdot |\{1, 2\}| = 16$ configurations (Fig. 8 reports measured controller time).

Proposition 1 (Controller guarantee, idealised). *Consider the idealised system with equal-size clients, a fixed number m of selected clients and a stationary utility process, and assume a Slater condition (there is a randomised policy meeting all queue constraints with slack $\varsigma > 0$). Let U^* be the optimal time-average utility of the relaxed problem and let the shadow price be absorbed into the utility, $U' = U - (\lambda/V) e/\bar{E}$. Then the policy that maximises (13)*

each round satisfies $\liminf_T \frac{1}{T} \sum_t \mathbb{E} U'_t \geq U'^* - \frac{C_0}{V}$ and $\limsup_T \frac{1}{T} \sum_t \mathbb{E} [q_k^t + \sum_j Z_j^t] \leq \frac{C_0 + V(\bar{U} - U'^*)}{\varsigma}$, with C_0 a constant depending only on the second moments of the arrivals.

This is the standard drift-plus-penalty trade-off (Neely, 2010): larger V approaches the optimal utility at the price of larger queues, i.e. larger transient budget violations. The proposition covers the idealised system only (the practical controller has unequal client sizes, a coverage reward linearised as $1/m$, a hard battery, and a utility that depends on stale loss statistics); we therefore evaluate the constraints empirically in Section 6. We also note that the shadow price λ is not part of the stability argument—it only changes the utility to U' —and is tuned on validation data like V .

Aggregation and quantisation in the implementation. Block l of the new global model is $\mathbf{w}_l^{t+1} = \mathbf{w}_l^t + \sum_{k: d_k \geq l} n_k \Delta_{k,l} / \sum_{k: d_k \geq l} n_k$, where $\Delta_{k,l}$ is the (possibly quantised) update. Quantisation is per-tensor stochastic uniform rounding to 2^b levels, which is unbiased. Blocks that no selected client trained keep their value.

5. Experimental setup

5.1. Datasets and federated partitions

We use three public benchmarks from two sensing domains whose natural unit of heterogeneity is the person: two wearable inertial datasets and one audio dataset. In all three, training, validation and test sets are separated *by person* (subject or speaker); inertial windows overlap by 50%, so any window-level split would leak between training and test.

UCI HAR (Anguita et al., 2013): 30 volunteers, six activities, nine inertial channels (body acceleration, total acceleration, gyroscope) in windows of 128 samples at 50 Hz. We keep the official split. Of the 21 training subjects, five are held out for validation (threshold calibration and hyper-parameter selection) and each of the remaining 16 is one federated client. The test set is the 9 official test subjects (2947 windows).

PAMAP2 (Reiss and Stricker, 2012): nine subjects, 12 protocol activities, three body-worn IMUs (hand, chest, ankle). We use the accelerometer ($\pm 16g$) and gyroscope of each IMU (18 channels), resample from 100 to 50 Hz and cut windows of 128 samples with 50% overlap, labelled by the majority activity and dropped when the label is transient or less than 90% pure. Subject 109

has only 48 windows and is excluded. Each of the remaining eight subjects is split into four contiguous time chunks; a client is a (subject, chunk) pair. Because the protocol is sequential, a chunk covers only three to five activities, which gives a strongly non-IID partition (Fig. B.9). We rotate four subject folds: in fold f two subjects are the test set, one is validation and five subjects (20 clients) train. Seeds are assigned to folds cyclically so that every method sees the same fold for the same seed.

Speech Commands v0.02 (Warden, 2018): 105 829 one-second clips of 35 spoken words from 2 618 speakers. We use the standard 12-class keyword-spotting task: ten keywords (yes, no, up, down, left, right, on, off, stop, go), an “unknown” class made of the other 25 words (capped at 25% of a client’s or evaluation set’s clips) and a “silence” class (10%) made of one-second crops of the background-noise recordings, with different noise files used for training, validation and test. Each clip is converted to a 40-bin log-mel spectrogram with 48 frames (32 ms window, 20 ms hop), which the one-dimensional network treats as 40 channels over 48 time steps. The dataset’s own validation and test lists are speaker-disjoint from the training speakers; the test set has 6 268 clips and the validation set 1 667. Each federated client is one of the 24 training speakers with the most clips (71–184 clips each, 2 571 in total, mean 107), so the partition is by speaker; label skew is mild because every speaker recorded a random subset of words, whereas the speaker (voice, microphone, room) shifts the features. Compared with the wearable datasets, the clients are small and the task has more classes, so accuracies are lower and learning is slower.

5.2. Model and federated protocol

The network has $L = 4$ blocks (Conv1d with kernel 5, GroupNorm, ReLU, max-pool) with 16, 32, 48 and 64 channels and a linear head after global average pooling at each block. On the 9-channel UCI HAR input it has 27 784 parameters and 0.75 M MACs per forward pass through all blocks (per-block parameters 870/2 854/8 118/15 942); the first layer grows with the number of input channels (18 for PAMAP2, 40 for Speech Commands). We run $T = 80$ rounds, select $m = 5$ clients per round, use batch size 32, at most $\tau_{\max} = 2$ local epochs, plain SGD, and the configuration set $\{1, 2, 3, 4\} \times \{8, 32\} \times \{1, 2\}$ ($|\mathcal{C}| = 16$). The loss is the mean cross-entropy over the exits a client trains. Device classes are assigned to clients at random in the 40/40/20% proportion of Table 3. We test three budget tightnesses, $\phi \in \{0.1, 0.25, 1.0\}$: the fleet

capacity $\mathcal{C}$ equals about 8, 20 and 80 rounds of the nominal full-configuration spend, so at $\phi = 0.1$ energy is the binding resource, at $\phi = 1$ it is almost not.

5.3. Methods compared

FedAvg (McMahan et al., 2017) random selection among feasible clients, full depth, 32-bit upload, $\tau_{\max}$ epochs.

FedProx (Li et al., 2020a) FedAvg with a proximal term of weight μ .

FedAvg-Q8 FedAvg with 8-bit stochastic quantisation (Alistarh et al., 2017) (communication-only saving).

FedAvg-Paced FedAvg that admits a client only while its cumulative energy stays within its budget (plus one burst).

Static-Depth capability-class heterogeneity in the style of Diao et al. (2021); Kim et al. (2023); Ilhan et al. (2023): each client gets the deepest fixed depth that fits its budget at the nominal selection rate. This is our own simplified implementation of the idea (without the self-distillation of DepthFL or the width scaling of HeteroFL), not a reproduction of those systems.

Oort-style (Lai et al., 2021) top- m by statistical utility times a speed penalty, with an exploration fraction ϵ (simplified: no pacer, no staleness).

Energy-greedy top- m by statistical utility per Joule, with exploration.

EcoFed Section 4.2, and ablations that remove the energy queue (*noQ*), the coverage queue (*noZ*), depth control (*fixdepth*), precision control (*fixprec*) or epoch control (*fixtau*).

All baselines are battery-aware (a client cannot be picked if it cannot afford the configuration) so that no method is penalised merely for being energy-blind; energy-blind variants would lose even more.

5.4. Tuning protocol

Hyper-parameters are chosen on *validation data only*, on two tuning seeds (100, 101) that are disjoint from the evaluation seeds, separately for every dataset and for $\phi \in \{0.1, 0.25\}$. **Every method receives exactly 14 trials.** For each method the search space is the product of its own knobs and a common grid of 16 learning rates (geometric from 0.02 to 0.8), and 14 distinct points are drawn at random without replacement with a seed that depends only on the dataset, budget and method. The knobs are:

FedProx $\mu \in \{10^{-3}, 10^{-2}, 10^{-1}\}$; Oort-style exploration $\epsilon \in \{0.1, 0.3\}$ and speed exponent $\xi \in \{1, 2\}$; Energy-greedy $\epsilon \in \{0.1, 0.3\}$; EcoFed $\rho^{\min} \in \{0.3, 0.6\}$, $\lambda \in \{0.03, 0.1, 0.3\}$ and $\kappa_d \in \{1, \frac{1}{2}, \frac{1}{4}\}$ with $V = 2$ (a space of $18 \times 16 = 288$ points of which EcoFed samples 14, whereas FedAvg samples 14 of 16). Methods with larger spaces are therefore searched more sparsely, which is a handicap for EcoFed rather than an advantage. The criterion is the area under the best-exit validation accuracy as a function of the fraction of fleet capacity spent. The setting $\phi = 1$ reuses the configurations tuned for $\phi = 0.25$ to limit computation. Single-exit FedAvg uses FedAvg’s tuned settings.

5.5. Metrics and statistics

Best-exit accuracy. A depth-heterogeneous method may leave deep exits poorly trained, so we report test accuracy at the exit chosen by validation accuracy (“best exit”); the last-exit accuracy is reported separately. **Final accuracy:** best-exit accuracy after 80 rounds (mean of the last three evaluations). **Best checkpoint:** the evaluation point with the highest smoothed best-exit validation accuracy; we report its test accuracy and the fleet energy spent up to it. This is an early-stopping oracle that every method gets. **Accuracy at an energy fraction $A@f$:** test accuracy once the fleet has spent a fraction $f \in \{0.1, 0.25, 0.5, 1\}$ of its capacity (methods that stop earlier keep their final accuracy). **Energy to target (ETA):** the fleet energy at which the smoothed best-exit test accuracy first reaches 90% of the accuracy FedAvg reaches at $\phi = 1$ (45.8% on PAMAP2, 87.0% on UCI HAR, 56.4% on Speech Commands; mean over seeds); runs that never reach it are censored, and ratios between methods use the seeds in which both reach it. **Battery depletion:** the fraction of clients whose battery is below 2% of their capacity at the end. **Coverage:** realised fraction of selected samples that train block j . All comparisons are paired by seed. We report mean differences with 95% bootstrap confidence intervals and two-sided paired t -tests, Holm-corrected within each (dataset, ϕ) setting across the seven baselines. There are 16 evaluation seeds (0–15) per method and setting for the main comparison and 8 (0–7) for ablations, the shadow-price sweep and the sensitivity studies. On PAMAP2 the four subject folds are cycled by seed, so that each fold is used 4 times.

5.6. Implementation and safeguards

The simulator is deterministic given the seed: re-running a job reproduces every logged number exactly, which we verified by re-running three jobs under a different experiment name. All data splits are by subject; validation subjects are used for hyper-parameter selection, for the choice of the best exit and checkpoint, and for the calibration of the cascade threshold, never for training, and the test subjects are used only to report numbers. Normalisation statistics are computed on training clients only. Local training uses plain SGD without momentum, batch size 32, no weight decay, and GroupNorm, so that model averaging is well defined. The 80-round horizon was fixed before any experiment. Evaluation is performed after every round for the first 24 rounds, when energy is spent fastest, and every fourth round afterwards. The experiments (several thousand federated runs including tuning, sweeps, sensitivity studies, pilots and abandoned tuning rounds) were executed on four CPU cores; a single run takes between 15 and 150 s. The decisions that were made after looking at some results, all on tuning seeds and validation data, were the exit weighting (Appendix B) and the evaluation of the best exit instead of the last exit. A first version of this study tuned the methods unequally (EcoFed 14 trials, the baselines 3 to 6) on two datasets and evaluated 16 seeds; because unequal budgets favour the method with the most tuning, the tuning was redone with equal budgets, a third dataset (Speech Commands) was added, and *all* results in this paper come from the second, equal-budget study; the first study’s numbers are not reported. The shadow-price sweep, the coverage probe, the sensitivity studies and the overhead check are reported in full, not as selected cases.

6. Results

All statements below refer to the evaluation seeds (16 per method and setting for the main comparison, 8 for ablations and sensitivity studies), to test subjects that never appear in training or validation, and to *model-based* energy (Section 3.2). “Significant” means a Holm-corrected paired t -test at $p < 0.05$ within a (dataset, ϕ) setting across the seven baselines; the full paired tables are in Appendix B.3. Differences are in percentage points (pp) of test accuracy.

6.1. Accuracy under a fixed energy budget

Table 4 and Fig. 1 compare all methods. The outcome depends on the dataset, and we report it per dataset rather than as an average.

PAMAP2 (strong label skew): large gains at binding budgets. At $\phi = 0.25$ EcoFed’s final accuracy is 57.5%, against 29.2–42.8% for the baselines; the paired differences are +15.0 to +28.3 pp and significant against all seven baselines, including Static-Depth (+15.0 pp, 95% CI [11.6, 18.3]). At the validation-selected checkpoint it reaches 57.0% against 40.1–47.7% for the six energy-blind baselines (+9.3 to +17.0 pp, significant), and 55.9% for Static-Depth (+1.1 pp, CI [−3.3, 5.3], not significant). At $\phi = 0.1$ the final accuracy (42.0%) is again significantly higher than that of every baseline (+4.4 to +16.2 pp), but with early stopping Static-Depth is clearly better: 54.0% against 41.5% (−12.5 pp, CI [−17.9, −7.0], Holm $p = 0.004$), whereas EcoFed’s advantage over the six other baselines at the checkpoint (+2.6 to +6.9 pp) is not significant after correction. Static-Depth’s final accuracy is 37.6% and its last exit is essentially untrained (9.2% accuracy), so it is the best choice at the tightest budget only if the operator can detect the peak and stop and does not need deep exits. At $\phi = 1$ EcoFed’s final accuracy (57.6%) is tied with Static-Depth (57.7%) and FedAvg-Paced (57.6%) and significantly higher than that of the five others (+5.7 to +11.8 pp); at the checkpoint no difference is significant (EcoFed 54.8%, others 52.4–59.8%).

UCI HAR (near-IID labels): equal accuracy for less energy. In UCI HAR every client sees all six activities, only the person and the sensor placement change, and all methods reach 84–89% within a few rounds (Fig. 1, top). At the validation-selected checkpoint EcoFed is statistically indistinguishable from all baselines in all three budgets (differences between −1.1 and +0.8 pp). Its final accuracy is significantly higher than every baseline at $\phi = 0.1$ (+1.8 to +3.2 pp: energy-blind methods lose accuracy once their batteries run out), higher than FedAvg-Q8, FedProx and FedAvg by 2.5–3.3 pp at $\phi = 0.25$, and tied at $\phi = 1$. The saving is in energy: over the run EcoFed uses 2.0 times less fleet energy than FedAvg at $\phi = 1$ (187 J against 379 J) and 1.15 times less at $\phi = 0.25$, and only 0.4% of devices end below 2% battery against 16–31% for the baselines other than Static-Depth (4.7%). At $\phi = 0.1$ the baselines stop early because remaining batteries cannot afford a full-depth round and leave 28% of the capacity stranded, whereas EcoFed uses 77% of it and ends with 14% of devices drained.

Speech Commands (small clients, 12 classes): EcoFed is not the best method. Here the result is mixed to negative. At $\phi = 0.1$ EcoFed’s final accuracy is higher than that of every baseline (+1.6 to +4.7 pp; significant only against FedAvg) and its checkpoint accuracy is tied with all of them except FedAvg (+5.9 pp). At $\phi = 0.25$ EcoFed is better than Oort-style, Energy-greedy and FedAvg in final accuracy (+2.7 to +3.2 pp) but worse than FedAvg-Paced (−6.9 pp, CI [−8.9, −4.9]) and, at the checkpoint, worse than FedAvg-Paced (−7.8 pp), FedAvg-Q8 (−3.2 pp) and Static-Depth (−2.6 pp). At $\phi = 1$ it is significantly worse than all seven baselines at the checkpoint (−2.5 to −6.1 pp) and than four of them in final accuracy. EcoFed spends roughly half the energy of the others on this dataset (12 against 20 J at $\phi = 0.25$, 51 against 95 J at $\phi = 1$) and is frugal by configuration: the shadow price λ selected at $\phi = 0.25$ is reused at $\phi = 1$, where it is too high (Section 6.5), but even with the lowest price in our sweep EcoFed does not reach the accuracy of the best baselines at $\phi = 1$. We do not have a tested explanation. A plausible one is that with only 71–184 clips per client and a speaker-level feature shift, deeper local training and more epochs matter more than cheap breadth of participation.

6.2. Energy to reach a target accuracy

Fig. 2 reports the fleet energy needed to reach 90% of the accuracy that FedAvg reaches when energy is not scarce. On UCI HAR EcoFed needs 6.5, 7.3 and 13.9 J at $\phi = 0.1$, 0.25 and 1, which is 1.7 to 4.2 times less than FedAvg, FedProx, FedAvg-Q8 and FedAvg-Paced (all significant), 2.2 to 2.5 times less than Oort-style and Energy-greedy at $\phi \leq 0.25$, and the same as those two at $\phi = 1$ (ratios 1.19 and 0.98, not significant). Static-Depth is tied with EcoFed at $\phi = 0.1$ (ratio 0.96) and slower at $\phi = 0.25$ and 1 (1.5 and 1.7 times, significant). On PAMAP2 the target is not always reached within the horizon at $\phi = 0.1$ (9 of 16 seeds for EcoFed, 15 of 16 for Static-Depth, 4 to 7 of 16 for the others), so ratios there use the seeds in which both reach it and are not significant; at $\phi = 0.25$ EcoFed needs 28 J, 2.1 to 3.8 times less than all baselines except Static-Depth (ratio 1.04, tied), and at $\phi = 1$ (60 J) it is 1.7 to 2.6 times faster than Oort-style, FedAvg-Paced, FedAvg-Q8, FedProx and FedAvg, and tied with Energy-greedy (1.19) and Static-Depth (1.15). On Speech Commands no method reaches the target at $\phi = 0.1$, and at $\phi = 0.25$ only FedAvg-Paced (9 of 16 seeds), Static-Depth (2) and FedAvg-Q8 (1) do, EcoFed never; at $\phi = 1$ EcoFed (35 J) is tied with Oort-style, Energy-greedy

and FedAvg-Q8 (ratios 1.03, 0.94, 1.07) and 1.3 to 1.5 times faster than FedAvg-Paced, FedProx, FedAvg and Static-Depth.

6.3. Where the gains come from: ablations

Table 5 removes one ingredient at a time (8 seeds). Differences are paired by seed; p -values below are unadjusted paired t -tests and should be read as descriptive.

The energy queue is essential. Without it clients keep running expensive configurations until their batteries are empty: 15 to 72 pp more devices end drained (all six settings, $p \leq 0.002$), up to 38 J more energy is spent, and final accuracy drops by 7.8 and 6.5 pp on UCI HAR ($p \leq 0.010$), by 8.2 ($p = 0.14$) and 27.8 pp ($p < 0.001$) on PAMAP2 and by 2.1 ($p = 0.023$) and 8.1 pp ($p < 0.001$) on Speech Commands, at $\phi = 0.1$ and 0.25 respectively. *Depth control matters most, then epoch control.* Forcing full depth lowers final accuracy by 4.5 to 18.0 pp in every setting ($p \leq 0.042$) while using up to 97 J less energy because clients cannot afford the configuration. Forcing two epochs changes final accuracy by +0.7 to −6.9 pp and is significant in two of six settings (PAMAP2, −6.9 and −3.1 pp). *Precision control matters little here.* Forcing 32-bit uploads changes final accuracy by at most 1.8 pp and is never significant; this is expected given the energy breakdown of Fig. 3, where radio energy is a small share of the cost of a participation in our default setting. We return to this in Section 6.6. *The coverage queue protects the deep exits, at a price.* Without it the last exit loses 10.2 pp on UCI HAR at $\phi = 0.1$ (not significant), 29.0 pp at $\phi = 0.25$ ($p = 0.040$), 21.5 and 22.9 pp on PAMAP2 ($p \leq 0.003$) and 30.2 pp on Speech Commands at $\phi = 0.25$ ($p < 0.001$); at $\phi = 0.1$ on Speech Commands there is no effect (+0.9 pp). The best-checkpoint accuracy is not significantly changed in any setting, because shallow exits are enough to classify; and the final best-exit accuracy changes by −0.1 to −2.3 pp except on PAMAP2 at $\phi = 0.1$ (−13.2 pp, $p = 0.001$) and on Speech Commands at $\phi = 0.1$, where removing the queue *improves* it by 4.7 pp ($p = 0.008$). The queue also costs energy: without it fleets spend 4 to 57 J less at $\phi = 0.25$. So the coverage queue buys the quality of the whole multi-exit model, not of its best exit, and on the small Speech Commands clients at the tightest budget it is a net cost. This is the behaviour the theory predicts for the deep exits, and we test it directly next.

6.4. The coverage floor is visible in the data

Theorem 1 predicts that, for a fixed horizon, the final gradient norm grows with $(1 - \sigma)^2$. We test this with a controlled probe that removes every other source of variation: no energy limit, uniform client sampling, and a depth that is $L = 4$ with probability σ and 1 otherwise, so that $\Pr[d \geq j] = \sigma$ for $j \geq 2$ (6 seeds $\times$ 8 values of $\sigma \times$ 2 datasets, 100 rounds). Fig. 4 shows the squared norm of the gradient of the full multi-exit objective at the final iterate. It rises with $(1 - \sigma)^2$ on both datasets: the linear fit has slope 0.88 and $R^2 = 0.14$ on UCI HAR (Spearman $\rho = 0.30$, $p = 0.037$) and slope 18.7 and $R^2 = 0.24$ on PAMAP2 ($\rho = 0.51$, $p = 2 \times 10^{-4}$). The relation is noisy and the bound is not tight—we do not claim it is—but the sign, the monotone trend and the quadratic shape are what the theorem predicts, and the last-exit accuracy falls accordingly (from 90.3% at $\sigma = 1$ to 87.6% at $\sigma = 0.1$ on UCI HAR; Fig. 4, right). Note that the practical aggregator used in the experiments is the sample-normalised average, not the Horvitz–Thompson weighting of the analysis; the probe therefore tests the prediction for the aggregator that is actually deployed.

6.5. An energy–accuracy frontier from one knob

The shadow price λ of Eq. (13) moves EcoFed along an energy–accuracy frontier, whereas a baseline’s operating point is simply the one at which it stops. Fig. 5 shows the validation-selected checkpoint (accuracy against the fleet energy spent up to it) for four values of λ with all other settings fixed at their tuned values, on the first eight evaluation seeds; the sweep is reported in full and was not used to select anything, and the baselines in the figure are the same eight seeds.

UCI HAR. At $\phi = 1$, $\lambda = 0.3$ (the tuned value) reaches 88.4% with 100 J, within 0.3 pp of FedAvg-Q8 (88.6%, 138 J) and FedAvg-Paced (88.7%, 171 J) and equal to FedAvg (88.3%, 228 J), i.e. 27% to 56% less energy for essentially the same accuracy; $\lambda = 1$ gives 86.1% with 74 J. At $\phi = 0.25$ the curve is on top of the baselines: $\lambda = 0.03$ reaches 88.1% with 86 J, against 87.5% with 67 J for FedAvg and 87.8% with 87 J for FedAvg-Q8. **PAMAP2.** At $\phi = 0.25$ the tuned setting ($\lambda = 0.1$) reaches 57.8% with 141 J and lies above every baseline (best baseline: Static-Depth 54.7% with 108 J; best energy-blind baseline 48.1%). At $\phi = 1$ it does not: $\lambda = 0.03$ gives 56.3% with 446 J and the tuned $\lambda = 0.1$ gives 54.1% with 331 J, below Static-Depth (58.4%, 348 J) and FedAvg-Q8 (60.4%, 417 J). **Speech Commands.** At $\phi = 0.25$ the best EcoFed point is 48.8% with 19 J ($\lambda = 0.1$), against 53.2%

with 19 J for FedAvg-Paced and 47.5% with 19–23 J for FedAvg-Q8 and Static-Depth; at $\phi = 1$ the lowest price gives 59.5% with 84 J, 0.4 to 5 pp below the baselines (59.9–64.5% with 86–93 J).

In short, the frontier lies on or above the baselines on UCI HAR and on PAMAP2 at the binding budget, where it saves energy or accuracy, and below the best baselines at the loose budget on PAMAP2 and on Speech Commands. The lower accuracy of the tuned EcoFed at $\phi = 1$ in Table 4 is partly a matter of operating point: the price selected at $\phi = 0.25$ is too frugal for $\phi = 1$ (the sweep shows a higher accuracy at $\lambda = 0.03$ for PAMAP2 and Speech Commands). Retuning $\phi = 1$ separately is the obvious fix and would be fairer to EcoFed; we did not do it because of its computational cost.

6.6. Robustness to the energy model

Because energy is modelled, we ask how much the conclusions depend on the model (Fig. 6; PAMAP2, $\phi = 0.25$, 8 seeds, metric: area under test accuracy as a function of the energy fraction spent).

Radio-to-compute cost ratio. Multiplying all radio energies by 10 or 100 turns communication from a minor into a major cost. EcoFed’s area under the curve falls from 49.8% to 47.0% and 40.8% (a loss of 9.0 pp), while FedAvg, FedAvg-Q8, Oort-style and Energy-greedy fall from 32.9–35.9% to 18.9–23.1% (losses of 12.8 to 15.6 pp). Even FedAvg-Q8, which quantises uploads to 8 bits, loses 12.8 pp, because it still sends the entire network; EcoFed can reduce the bits sent by choosing a smaller depth as well as a smaller bit-width. So precision control, which has no measurable effect in the default setting (Section 6.3), becomes useful together with depth control when radio energy dominates. This experiment was run on one dataset only.

Wrong energy beliefs. We let the energy that actually drains the batteries differ from the controller’s belief by independent log-normal factors on the arithmetic and memory constants of every device ($\sigma = 0.25$ and 0.5). EcoFed’s area under the curve is 49.8%, 50.3% and 50.2% for $\sigma = 0, 0.25, 0.5$ (unchanged within the seed-to-seed variation), which is what one expects when queues integrate the energy that devices report rather than the energy that the model predicts. The other methods are also fairly insensitive: Static-Depth loses 1.7 pp (48.4 $\rightarrow$ 46.7%) and Oort-style 2.0 pp at $\sigma = 0.5$; Energy-greedy and FedAvg-Paced change by at most 0.6 pp. The baselines are battery-aware by construction, so this experiment shows that EcoFed is not *more* fragile than they are, rather than that it is more robust.

Unmodelled per-sample overhead. Section 6.9 shows that on a real CPU a participation costs a fixed amount per sample on top of the MAC-proportional part, equivalent to 0.55–0.75 million MACs. We add 0.5 and 1.0 million MAC-equivalents per training sample to the *true* energy only. On PAMAP2 EcoFed’s area under the curve falls from 49.8% to 47.1% and 45.1% (losses of 2.7 and 4.7 pp), Static-Depth’s from 48.4% to 44.2% and 40.8% (4.2 and 7.6 pp) and FedAvg-Paced’s from 36.5% to 34.8% and 34.0%; the energy-blind methods lose 1.7 to 2.3 pp at 0.5 million and 2.0 to 3.9 pp at 1.0 million. EcoFed stays ahead of every baseline, by 2.9 and 4.3 pp over Static-Depth. On UCI HAR all methods stay within 3 pp of each other: EcoFed’s area is 84.3% and 83.7%, FedAvg-Paced’s 84.3% and 85.3%, Static-Depth’s 82.2% and 80.9%. A cost model that overstates the benefit of shallow updates therefore reduces but does not remove the advantage of depth control on the dataset where it matters.

6.7. Lifecycle energy

Training is only half of the energy bill. We deploy each trained model with the confidence-gated cascade of Section 3.4: a single threshold θ per model is chosen on validation data as the cheapest one whose validation accuracy is within one percentage point of the model’s best single exit, and we then report test accuracy and modelled inference energy (fleet-average device mix). Table 6 compares three deployments: single-exit FedAvg (trained with a last-exit loss; every window runs the full network), multi-exit FedAvg with the cascade, and EcoFed with the cascade. For a fleet that classifies one window every 1.28 s per device (inertial data) or every 0.5 s (speech), always on, the last column gives the time after which the energy spent on serving exceeds the energy spent on training ($M_{\times}$ divided by the fleet’s window rate).

Four observations follow. *First, the cascade cuts inference energy per window by a factor of 1.3 to 7.2.* EcoFed’s cascade needs 118 and 130 μJ per window on UCI HAR against 845 μJ for the single-exit network (7.2 and 6.5 times less), at a cost of 0.7 and 1.2 pp of accuracy at $\phi = 0.25$ and 1; on PAMAP2 it needs 219 and 382 μJ against 950 μJ (4.3 and 2.5 times less) and is far more accurate than the single-exit model (58.5% and 57.9% against 22.1% and 39.9%; see the caveat on single-exit tuning in Section 8); on Speech Commands the saving is small (331 and 349 against 454 μJ , 1.4 and 1.3 times) because the 12-class task needs the deeper exits. The multi-exit FedAvg cascade saves less on the two wearable datasets (2.3 to 3.9 times) and almost nothing on Speech Commands (1.2 times), because its shallow exits are weak:

only 23–70% of windows leave at the first exit on the wearable datasets, against 70–97% for EcoFed, and none on Speech Commands. Table B.12 shows why: with exit weights $\lambda_j \propto j^2$ the first exit receives only 3.3% of the loss weight of a full-depth client, and FedAvg’s first exit reaches 51.5% on UCI HAR against 86.1% for EcoFed, whose depth-1 clients train it with the full loss weight. This is a consequence of our choice of exit weights (made to favour the last exit, Appendix B); with uniform weights the shallow exits of the energy-blind baselines would be better and their last exit worse, so the cascade comparison must be read with that choice in mind. *Second, the extra training cost of making a model multi-exit is negligible.* EcoFed trains with *less* energy than FedAvg in five of six settings; in the sixth (PAMAP2, $\phi = 0.25$) it uses 27 J more (163 against 136 J) but saves 731 μ J per window relative to the single-exit network, so by Eq. (5) the extra training is repaid after 3.7×10^4 windows, about 40 minutes of operation of the 20-device fleet. *Third, serving dominates the lifecycle.* In every row serving matches the training energy within 12 minutes to 1.7 days of fleet operation (last column of Table 6; Fig. 7), so over a month of operation inference would use roughly 18 to 3 400 times the energy of training. *Fourth, these numbers depend on the always-on assumption* (one window per 1.28 s per device for the inertial data and two per second for speech) and on modelled energies, and should be read as orders of magnitude.

6.8. Controller overhead

Controller cost. EcoFed scores $|\mathcal{C}| = 16$ configurations per client and sorts the best scores. Fig. 8a reports the measured time of the selection step in our (unoptimised, pure Python) implementation on synthetic fleets with 5% of clients selected: 3 ms for $N = 100$, 0.036 s for $N = 10^3$, 0.26 s for $N = 10^4$ and 2.7 s for $N = 10^5$, 2.6 times the cost of an Oort-style top- m ranking at $N = 10^5$ (1.0 s) and 8 to 14 times at $N \leq 10^4$. This is negligible compared with a federated round, which takes seconds to minutes at that scale, but it is a Python measurement of the selection step only and says nothing about the scalability of learning itself, which we did not test beyond 20 clients.

6.9. Cost model versus measured latency

The relative cost of depth is the lever behind depth control, so we checked it against wall-clock time. We timed one training step of the multi-exit network at depths 1–4 on a single CPU thread (batch 32, median of 40 repetitions). Measured time per sample grows more slowly than the modelled

cost: depth 4 takes 3.4 times as long as depth 1 on UCI HAR’s 9-channel input and 3.1 times on PAMAP2’s 18-channel input, while the MAC count grows by a factor of 8.1 and 4.6 (Fig. 8b). A linear model of time in MACs fits well ($R^2 = 0.96$ and 0.98) with a fixed cost per sample equivalent to about 0.75 and 0.55 million MACs. We therefore do not claim that the modelled savings from shallow updates are what a CPU would show; the sensitivity experiment of Section 6.6 injects that overhead into the true energy and shows that EcoFed’s advantage on PAMAP2 shrinks but persists. Latency is not energy, and an accelerator would behave differently; this check only validates the *relative* depth scaling on one CPU.

7. Discussion

What the evidence supports. Three findings are supported by paired statistics over 16 seeds, with equal tuning budgets for all methods. First, when the budget is binding and the labels are strongly skewed across clients (PAMAP2, $\phi \leq 0.25$), letting clients choose how much of the network to train, under energy prices and a coverage constraint, gives a much better final model than FedAvg-type training with random or utility-based selection (+4 to +28 pp in final accuracy, significant against every baseline) and, at $\phi = 0.25$, also a better early-stopped model (+9 to +17 pp against the six energy-blind baselines). Second, when labels are similar across clients (UCI HAR) the same controller reaches the same early-stopped accuracy as the baselines with up to 2 times less fleet energy, 1.7 to 4.2 times less energy to a target accuracy than FedAvg-type methods, and with far fewer drained devices. Third, the ingredients that matter are the energy queue, depth control and (less so) epoch control; the coverage queue protects the accuracy of the deep exits; precision control matters only when radio energy is expensive relative to computation (Section 6.6). The convergence bound is consistent with the coverage probe, although the relation is noisy and the bound is loose.

What it does not support. EcoFed does not dominate, and the paper’s claims should be limited accordingly. (i) On Speech Commands, with small clients and a 12-class task, EcoFed is worse than several baselines in early-stopped accuracy at $\phi = 0.25$ and worse than all seven at $\phi = 1$, by 2.5 to 7.8 pp. (ii) A simple capability-based depth rule (Static-Depth) reaches the best early-stopped accuracy on PAMAP2 at the tightest budget (+12.5 pp over EcoFed) and ties EcoFed at the checkpoint at $\phi = 0.25$ and at $\phi = 1$; its

energy to a target accuracy is statistically indistinguishable from EcoFed’s on PAMAP2. What EcoFed adds over that rule is accuracy at the *end* of the run, usable deep exits, protection of the batteries and a single knob, λ , that trades energy for accuracy (Section 6.5). (iii) At loose budgets ($\phi = 1$) on PAMAP2 and Speech Commands the tuned EcoFed is partly held back by a shadow price that was tuned at $\phi = 0.25$ and reused, so those rows are a conservative picture of the method. (iv) On UCI HAR the most accurate model is single-exit FedAvg; a multi-exit model is worth its small extra training cost only if the deployment-time savings of Section 6.7 are wanted.

Why depth control may help when labels are skewed. A plausible explanation, which we did not test in isolation, is breadth of participation: cheap shallow updates let many clients—and therefore many activities—contribute in every phase of the run, while full-depth updates exhaust a few clients and then let the fleet shrink to those that can afford them. The collapse of the energy-blind baselines after their peak on PAMAP2 (Fig. 1), the near-parity on UCI HAR, where every client holds every label, and the weaker result on Speech Commands, whose clients are small and not label-skewed, are all consistent with this reading.

Implications for sustainable edge AI. The lifecycle analysis reframes where the energy goes. In our model, a fleet that serves a continuous classifier spends as much energy on inference as it spent on training within 12 minutes to 1.7 days, and over a month 18 to 3 400 times as much (Table 6). Two consequences follow. A federated training controller should be judged not only by the Joules it saves during training—which in absolute terms are small—but by whether it produces models that can be served cheaply: EcoFed’s shallow exits are accurate enough that 70–97% of windows leave at the first block on the wearable datasets, while the exits of a model trained with plain multi-exit FedAvg are not (23–70%). And the training cost of making a model multi-exit is repaid within minutes to hours of operation, so there is no energy argument for deploying a single-exit network, even where single-exit training reaches a slightly higher accuracy (UCI HAR). On Speech Commands the deployment saving is small, because the task needs the deeper exits.

Practical guidance. (i) Estimate the budget regime first: if the energy per participation of the full model is within a factor of a few of the per-round budget and the data are label-skewed across clients, depth and epoch control pay off; if energy is abundant they mostly save energy. (ii) If the data

are close to IID across clients, expect little accuracy to be lost by shallow, cheap updates, and use the savings. (iii) If the operator can stop training at a validation-selected checkpoint and does not need deep exits, a static capability-based depth is a strong, simple baseline. (iv) Tune the shadow price per budget: it is the knob that sets the operating point. (v) Keep the energy accounting closed-loop: EcoFed’s queues use energy reported by the devices, which is what makes it indifferent to an incorrect energy model.

8. Limitations and threats to validity

- **Energy is modelled, not measured.** The device constants are assumptions and the paper contains no power measurement. Section 6.6 shows that conclusions are robust to radio cost, to multiplicative errors of up to 50% (log-normal $\sigma = 0.5$) in the controller’s belief, and to an unmodelled per-sample overhead fitted from CPU latency; but a power-metered Raspberry Pi/Jetson study is the necessary next step (the measurement and fitting harness is provided, Appendix B.6, but was not run), and the absolute Joules reported here should not be quoted as measurements.
- **Scale.** The fleets have 16–24 clients and the network 27 800–31 300 parameters; the selection is 5 clients per round over 80 rounds. We do not know how the ranking of methods changes with thousands of clients, deeper networks, or larger models; the controller overhead measurements (Section 6.8) say only that the algorithm itself scales.
- **Three datasets, two domains.** Two wearable inertial benchmarks and one keyword-spotting benchmark; vision and language, and large-scale cross-device settings, are not covered. The Speech Commands result is negative for EcoFed.
- **Equal trial counts, unequal spaces.** All methods received 14 random-search trials, but spaces differ: EcoFed samples 14 of 288 points and FedAvg 14 of 16, so EcoFed’s tuned configuration is likely further from its optimum. Several selected learning rates lie near the edge of the grid, so all methods may be slightly under-tuned. The setting $\phi = 1$ reuses the $\phi = 0.25$ configuration for every method; this probably hurts EcoFed most, because its shadow price depends on the budget.
- **Single-exit FedAvg.** It uses FedAvg’s tuned settings and was not tuned separately. Its comparison (UCI HAR: best; PAMAP2: poor) is indicative only.

- **Theory versus practice.** Theorem 1 assumes iterate-independent participation, bounded gradients and an unbiased Horvitz–Thompson aggregator; the experiments use utility-based selection and sample-normalised block averages. Proposition 1 is stated for an idealised system. The coverage experiment supports the qualitative prediction only.
- **Exit weights.** We weight exits proportionally to j^2 for all methods. This favours the last exit and handicaps the shallow exits of full-depth methods, which affects the cascade comparison (Table B.12); we did not test other weightings beyond the pilot of Appendix B.
- **Baselines.** Static-Depth and Oort-style are our simplified implementations of ideas from the literature, not reproductions of DepthFL/HeteroFL/ScaleFL or the Oort system.
- **Validation-selected checkpoint.** The best-checkpoint metric assumes a small validation set at the server; the final-accuracy metric does not.
- **Not modelled.** Stragglers and dropouts, wireless channel dynamics, charging, privacy mechanisms and security, and the energy of the server.

9. Conclusion

We studied federated learning of multi-exit networks on battery-limited fleets, where energy is spent twice—when training and when serving. A convergence bound shows that letting weak devices train only shallow blocks creates an error floor controlled by how often each block is trained; EcoFed turns this into a coverage constraint and combines it with per-device energy queues and a shadow price in a controller that costs $O(N|\mathcal{C}|)$ per round. Across three public benchmarks, with equal tuning budgets and 16 paired seeds, the picture is regime-dependent. On label-skewed PAMAP2 at a binding budget EcoFed improves the final accuracy of the federated model substantially; on near-IID UCI HAR it matches the accuracy of simpler methods at much lower energy and protects device batteries; on Speech Commands it is worse than several baselines, and a simple static depth rule beats it at the tightest budget on PAMAP2 when training can be stopped at the best checkpoint. The advantage grows when radio energy is expensive, and the controller degrades gracefully when its energy model is wrong. The results therefore support EcoFed as an energy-saving and battery-protecting controller whose accuracy benefits are strongest under label skew and binding

budgets, not as a universally better federated learner, and every energy figure is model-based. Future work should (i) calibrate the energy model on metered devices and repeat the study on hardware, (ii) retune per budget and scale to larger fleets and models, (iii) learn the utility of depth online instead of tuning its exponent, and (iv) couple the training-time controller with the deployment-time exit policy so that the energy of both phases is optimised together.

Declarations

Data and code availability. Both datasets are public (UCI HAR and PAMAP2, UCI Machine Learning Repository). All code, configurations, tuned hyper-parameters and raw result logs are in the accompanying repository.

Use of generative AI. A generative AI assistant was used to write and run the experiment code and to draft and edit the text; the authors are responsible for verifying every claim, number and reference.

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Appendix A. Proof of Theorem 1

Write $\gamma = \eta\tau$ and let $\hat{g}_t$ be the aggregated direction, so that $\mathbf{w}^{t+1} = \mathbf{w}^t - \gamma\hat{g}_t$. All expectations are conditional on $\mathbf{w}^t$ unless stated otherwise; the participation and depth process is independent of the iterate by assumption.

Step 1: bias decomposition. For client k with depth d , let $g_k = \frac{1}{\tau} \sum_{i < \tau} \tilde{\nabla} F_k^{(d)}(\mathbf{w}_{k,i})$ be the average of its local stochastic gradients along the local trajectory $\mathbf{w}_{k,0} = \mathbf{w}^t$. By (A2) $\|\mathbf{w}_{k,i} - \mathbf{w}^t\| \leq \eta i G$. Since stochastic gradients are unbiased given the local iterate and $F_k^{(d)}$ is L_s -smooth,

$$\|\mathbb{E}[g_k] - \nabla F_k^{(d)}(\mathbf{w}^t)\| \leq \frac{1}{\tau} \sum_{i < \tau} L_s \eta i G = \frac{L_s \eta G (\tau - 1)}{2} \leq \frac{L_s G \gamma}{2} =: \varepsilon.$$

Quantisation is unbiased (A3), so it does not change the mean. For block l the aggregate with weights $p_k/\pi_{k,l}$ has mean

$$\mathbb{E}[\hat{g}_{t,l}] = \sum_k p_k \mathbb{E}_d[\nabla_l F_k^{(d)}(\mathbf{w}^t) \mid d \geq l, k \in \mathcal{S}] + \text{drift}_l,$$

and by Minkowski's inequality the stacked drift over all blocks has norm at most ε .

Step 2: truncation bias. Since $\nabla_l F_k^{(d)} = \sum_{j=l}^d \lambda_j \nabla_l f_k^j$,

$$\mathbb{E}_d[\nabla_l F_k^{(d)} \mid d \geq l] = \sum_{j \geq l} \lambda_j \Pr[d \geq j \mid d \geq l] \nabla_l f_k^j.$$

For $j > l$ we have $\{d \geq j\} \subseteq \{d \geq l\}$, hence $\Pr[d \geq j \mid d \geq l] = \sigma_{k,j}/\sigma_{k,l} \geq \sigma_{k,j} \geq \sigma_j$, while the $j = l$ term has probability one. Comparing with $\nabla_l F_k = \sum_{j \geq l} \lambda_j \nabla_l f_k^j$,

$$\left\| \nabla_l F_k - \mathbb{E}_d[\nabla_l F_k^{(d)} \mid d \geq l] \right\| \leq G_l \sum_{j > l} \lambda_j (1 - \sigma_j),$$

and averaging over k with weights p_k gives block bias at most B_l . Stacking the blocks, $\mathbb{E}[\hat{g}_t] = \nabla F(\mathbf{w}^t) - b_t + e_t$ with $\|b_t\| \leq B$ and $\|e_t\| \leq \varepsilon$.

Step 3: descent. By L_s -smoothness of F , $F(\mathbf{w}^{t+1}) \leq F(\mathbf{w}^t) - \gamma \langle \nabla F(\mathbf{w}^t), \hat{g}_t \rangle + \frac{L_s \gamma^2}{2} \|\hat{g}_t\|^2$. Using $\langle a, a - c \rangle \geq \frac{1}{2} \|a\|^2 - \frac{1}{2} \|c\|^2$,

$$\mathbb{E} \langle \nabla F, \hat{g}_t \rangle \geq \frac{1}{2} \|\nabla F\|^2 - \frac{1}{2} (B + \varepsilon)^2,$$

and by (A4) $\mathbb{E} \|\hat{g}_t\|^2 \leq (\|\nabla F\| + B + \varepsilon)^2 + \sigma_h^2 \leq 2 \|\nabla F\|^2 + 2(B + \varepsilon)^2 + \sigma_h^2$. Therefore

$$\mathbb{E} F(\mathbf{w}^{t+1}) \leq F(\mathbf{w}^t) - \gamma \left(\frac{1}{2} - L_s \gamma \right) \|\nabla F\|^2 + \gamma \left(\frac{1}{2} + L_s \gamma \right) (B + \varepsilon)^2 + \frac{L_s \gamma^2}{2} \sigma_h^2.$$

For $\gamma \leq 1/(4L_s)$ the coefficients are at least $\frac{1}{4}$ and at most $\frac{3}{4}$. Taking total expectation, summing over $t < T$ and dividing by $\gamma T/4$,

$$\frac{1}{T} \sum_t \mathbb{E} \|\nabla F(\mathbf{w}^t)\|^2 \leq \frac{4\Delta}{\gamma T} + 3(B + \varepsilon)^2 + 2L_s \gamma \sigma_h^2.$$

Step 4: constants. Using $(B + \varepsilon)^2 \leq 2B^2 + 2\varepsilon^2$ and $\varepsilon \leq L_s G \gamma / 2$, we get $3(B + \varepsilon)^2 \leq 6B^2 + \frac{3}{2}L_s^2 G^2 \gamma^2 \leq 6B^2 + \frac{3}{8}L_s G^2 \gamma \leq 6B^2 + 2L_s \gamma \frac{G^2}{4}$, where the second inequality uses $\gamma \leq 1/(4L_s)$. Hence the bound is at most $\frac{4\Delta}{\gamma T} + 2L_s \gamma \tilde{\sigma}^2 + 6B^2$ with $\tilde{\sigma}^2 = \sigma_h^2 + G^2/4$. If $\gamma = \sqrt{\Delta/(L_s \tilde{\sigma}^2 T)} \leq 1/(4L_s)$ the first two terms sum to $6\sqrt{L_s \tilde{\sigma}^2 \Delta/T}$. Otherwise $\gamma = 1/(4L_s)$ and $\sqrt{\Delta/(L_s \tilde{\sigma}^2 T)} > 1/(4L_s)$, so the first term equals $16L_s \Delta/T$ and the second is at most $2\sqrt{L_s \tilde{\sigma}^2 \Delta/T}$. In both cases (8) holds. $\square$

Corollary 1.. With $\lambda_j = 1/L$, $G_l = G$ and $\sigma_j = \sigma$, $B_l = G(1 - \sigma)(L - l)/L$ and $B^2 = G^2(1 - \sigma)^2 \sum_{l=1}^L \left(\frac{L-l}{L}\right)^2 = G^2(1 - \sigma)^2 \frac{(L-1)(2L-1)}{6L}$, so $6B^2 = \frac{(L-1)(2L-1)}{L} G^2(1 - \sigma)^2$. Requiring this floor to be at most $\epsilon/2$ gives (9).

Appendix B. Additional material

Appendix B.1. Tuned hyper-parameters

Table B.7 lists the configurations selected on validation data. The $\phi = 1$ runs reuse the $\phi = 0.25$ configuration.

Appendix B.2. Client label distributions

Appendix B.3. Paired comparisons

Tables B.8–B.11 give EcoFed minus baseline (or baseline/EcoFed for ratios), paired by seed, with bootstrap 95% CIs and Holm-corrected p -values within each setting across the seven baselines; * marks $p < 0.05$.

Appendix B.4. Accuracy per exit

Appendix B.5. Pilot on exit weighting

In a pilot on tuning seeds (UCI HAR, FedAvg, 4 seeds) uniform exit weights gave a validation accuracy at the end of the budget of 81.9% at $\phi = 0.1$ and 87.5% at $\phi = 1$, whereas weights proportional to j^2 gave 85.1% and 88.5%, and a single-exit objective gave 86.4% and 88.6%; the single-exit model was clearly worse early in the run (68.9% against 79.6% at 25% of the capacity for the uniform weighting at $\phi = 0.1$). We therefore use $\lambda_j \propto j^2$ for all methods. Hyper-parameters were tuned after this decision.

Appendix B.6. Metered-hardware protocol (provided, not executed)

The study has no power measurements. To make the missing step cheap to carry out, the repository contains a harness that turns a metered device into a `DeviceProfile`: `energy/measure_hw.py` runs local training for every depth and for batch sizes 16, 32 and 64 on the device it executes on, brackets each window with timestamps, and integrates the power above the idle level from one of three sources (an external meter log such as Joulescope or Nordic PPK2, Jetson `tegrastats`, or an INA219 on a Raspberry Pi supply line); `energy/fit_profile.py` fits α_u , β_u and a fixed per-sample overhead c_0 by non-negative least squares to $E = \alpha \cdot 3 \text{MAC}_{1:d} + \beta \text{Bytes}_{1:d}(b) + c_0$ and reports R^2 and the mean absolute percentage error; radio energies per bit are obtained from timed transfers of files of known size over the real link. Setting the environment variable `ECOFED_DEVICE_JSON` replaces the assumed constants of Table 3 by the fitted ones in every experiment. The fitting code is covered by unit tests that recover known parameters from synthetic data (including with 5% noise); the three power-source readers are untested because the required hardware was not available, and no number in this paper depends on them.

Appendix B.7. Code and data

Everything needed to regenerate the tables and figures is in the repository: loaders for both datasets, the model, the energy model, the simulator, all methods, experiment configurations, tuned hyper-parameters and the analysis scripts. Raw per-run logs are regenerated by the experiment scripts (they are not committed because of their size).

Table 4: Main results (mean $\pm$ 95% CI over 16 seeds). Acc@25%: accuracy when 25% of the fleet capacity has been spent. Final acc.: accuracy of the best exit after 80 rounds. Best-ckpt: accuracy at the validation-selected checkpoint (an early-stopping oracle available to every method) and the fleet energy spent up to it. Energy used and the share of devices whose battery is below 2% of capacity (Depl.) at the end. Bold: best accuracy in the column within a setting.

Data / ϕ	Method	Acc@25%	Final acc.	Best-ckpt acc.	Ckpt energy (J)	Energy used (J)	Depl. (%)
UCI HAR, 0.1	EcoFed	83.4 $\pm$ 1.3	87.1 $\pm$ 0.9	86.4 $\pm$ 1.4	29	39	14.1
	Oort-style	75.2 $\pm$ 2.4	84.2 $\pm$ 1.5	86.1 $\pm$ 1.2	27	37	0.4
	Energy-greedy	79.0 $\pm$ 1.7	84.2 $\pm$ 1.6	86.9 $\pm$ 0.8	27	37	0.4
	Static-Depth	84.1 $\pm$ 0.9	85.0 $\pm$ 1.3	85.6 $\pm$ 1.3	25	49	36.7
	FedAvg-Paced	82.7 $\pm$ 1.9	84.2 $\pm$ 1.6	86.6 $\pm$ 1.0	23	37	0.4
	FedAvg-Q8	75.4 $\pm$ 4.6	83.9 $\pm$ 1.3	86.1 $\pm$ 1.2	28	37	8.6
	FedProx	77.7 $\pm$ 1.9	85.2 $\pm$ 1.3	86.6 $\pm$ 1.0	30	37	0.4
	FedAvg	72.5 $\pm$ 2.5	84.3 $\pm$ 2.3	87.5 $\pm$ 0.8	31	37	0.4
UCI HAR, 0.25	EcoFed	85.2 $\pm$ 1.1	87.4 $\pm$ 1.2	87.2 $\pm$ 1.4	69	101	9.0
	Oort-style	87.1 $\pm$ 1.0	85.6 $\pm$ 0.9	87.1 $\pm$ 1.0	70	116	16.8
	Energy-greedy	86.5 $\pm$ 0.9	85.4 $\pm$ 1.3	87.2 $\pm$ 1.0	72	116	16.8
	Static-Depth	84.8 $\pm$ 0.9	86.8 $\pm$ 0.8	86.9 $\pm$ 0.8	67	108	16.4
	FedAvg-Paced	84.6 $\pm$ 1.3	86.7 $\pm$ 1.0	86.7 $\pm$ 1.0	56	116	16.8
	FedAvg-Q8	85.5 $\pm$ 1.2	84.9 $\pm$ 1.1	87.6 $\pm$ 0.9	78	120	26.6
	FedProx	84.2 $\pm$ 1.6	84.5 $\pm$ 1.4	87.0 $\pm$ 0.8	73	116	16.8
	FedAvg	84.2 $\pm$ 1.6	84.1 $\pm$ 1.4	87.2 $\pm$ 0.9	70	116	16.8
UCI HAR, 1.0	EcoFed	87.3 $\pm$ 1.2	88.4 $\pm$ 1.0	87.8 $\pm$ 1.2	114	187	0.4
	Oort-style	88.2 $\pm$ 0.6	88.3 $\pm$ 0.8	88.1 $\pm$ 1.0	122	378	24.6
	Energy-greedy	87.9 $\pm$ 0.8	88.3 $\pm$ 0.7	88.2 $\pm$ 1.0	127	352	16.0
	Static-Depth	87.0 $\pm$ 1.4	88.7 $\pm$ 0.6	88.0 $\pm$ 0.8	190	270	4.7
	FedAvg-Paced	88.4 $\pm$ 0.7	89.1 $\pm$ 0.8	88.8 $\pm$ 0.7	212	389	24.6
	FedAvg-Q8	87.3 $\pm$ 0.8	87.5 $\pm$ 0.7	88.4 $\pm$ 0.5	162	353	31.2
	FedProx	87.7 $\pm$ 0.7	87.0 $\pm$ 1.0	88.1 $\pm$ 0.8	209	379	24.6
	FedAvg	87.4 $\pm$ 0.8	87.0 $\pm$ 1.0	88.0 $\pm$ 1.0	230	379	24.6
PAMAP2, 0.1	EcoFed	36.7 $\pm$ 4.1	42.0 $\pm$ 4.5	41.5 $\pm$ 6.1	21	32	0.6
	Oort-style	24.7 $\pm$ 1.8	25.8 $\pm$ 4.5	37.1 $\pm$ 3.8	38	47	5.6
	Energy-greedy	26.2 $\pm$ 3.3	29.4 $\pm$ 4.9	38.9 $\pm$ 4.5	40	47	5.6
	Static-Depth	40.5 $\pm$ 4.7	37.6 $\pm$ 5.4	54.0 $\pm$ 6.1	44	64	15.0
	FedAvg-Paced	22.3 $\pm$ 3.1	34.2 $\pm$ 4.6	36.3 $\pm$ 3.8	42	47	5.6
	FedAvg-Q8	25.4 $\pm$ 2.6	29.5 $\pm$ 3.8	34.7 $\pm$ 4.6	41	48	8.1
	FedProx	24.7 $\pm$ 2.1	28.5 $\pm$ 3.8	36.5 $\pm$ 4.6	38	47	5.6
	FedAvg	22.7 $\pm$ 1.9	29.3 $\pm$ 4.9	36.2 $\pm$ 5.2	46	47	5.6
PAMAP2, 0.25	EcoFed	50.8 $\pm$ 4.8	57.5 $\pm$ 3.6	57.0 $\pm$ 4.5	137	163	37.8
	Oort-style	36.8 $\pm$ 2.8	31.8 $\pm$ 4.9	46.4 $\pm$ 5.0	89	136	16.9
	Energy-greedy	41.6 $\pm$ 4.2	29.2 $\pm$ 4.0	47.7 $\pm$ 3.3	95	136	16.9
	Static-Depth	48.6 $\pm$ 4.2	42.5 $\pm$ 5.4	55.9 $\pm$ 5.3	116	172	34.7
	FedAvg-Paced	35.8 $\pm$ 3.5	42.8 $\pm$ 5.9	45.0 $\pm$ 6.2	115	136	16.9
	FedAvg-Q8	35.0 $\pm$ 3.0	34.7 $\pm$ 4.1	44.3 $\pm$ 5.8	98	136	15.3
	FedProx	33.2 $\pm$ 3.4	35.9 $\pm$ 5.4	40.9 $\pm$ 5.0	103	136	16.9
	FedAvg	33.1 $\pm$ 3.0	35.7 $\pm$ 5.6	40.1 $\pm$ 5.4	104	136	16.9
PAMAP2, 1.0	EcoFed	52.5 $\pm$ 4.9	57.6 $\pm$ 4.4	54.8 $\pm$ 4.9	285	414	4.1
	Oort-style	46.3 $\pm$ 4.6	46.4 $\pm$ 5.9	53.5 $\pm$ 5.9	303	561	20.3
	Energy-greedy	50.8 $\pm$ 4.8	51.9 $\pm$ 5.0	52.4 $\pm$ 5.2	331	518	8.4
	Static-Depth	53.8 $\pm$ 3.5	57.7 $\pm$ 3.8	58.8 $\pm$ 3.0	373	470	18.8
	FedAvg-Paced	51.7 $\pm$ 4.5	57.6 $\pm$ 4.4	57.3 $\pm$ 4.7	495	570	20.3
	FedAvg-Q8	46.7 $\pm$ 3.1	49.0 $\pm$ 6.4	59.8 $\pm$ 5.1	374	526	16.6
	FedProx	43.2 $\pm$ 3.1	48.0 $\pm$ 4.7	54.9 $\pm$ 4.9	367	556	20.3
	FedAvg	42.9 $\pm$ 2.7	45.8 $\pm$ 5.2	57.4 $\pm$ 4.8	412	556	20.3
Speech Commands, 0.1	EcoFed	31.0 $\pm$ 1.2	36.1 $\pm$ 1.2	36.3 $\pm$ 2.2	7	8	28.9
	Oort-style	30.5 $\pm$ 1.5	34.1 $\pm$ 0.6	34.1 $\pm$ 0.6	6	6	0.0
	Energy-greedy	30.6 $\pm$ 1.6	33.4 $\pm$ 1.5	32.9 $\pm$ 1.9	6	6	0.0
	Static-Depth	31.9 $\pm$ 1.2	32.8 $\pm$ 2.9	37.2 $\pm$ 2.2	7	8	7.0
	FedAvg-Paced	30.6 $\pm$ 2.1	34.2 $\pm$ 2.1	33.7 $\pm$ 2.5	6	6	0.0
	FedAvg-Q8	31.3 $\pm$ 1.1	34.5 $\pm$ 1.4	34.5 $\pm$ 1.7	6	6	3.4
	FedProx	30.6 $\pm$ 1.3	34.1 $\pm$ 1.1	33.5 $\pm$ 1.6	6	6	0.0
	FedAvg	31.3 $\pm$ 1.5	31.4 $\pm$ 2.3	30.4 $\pm$ 2.7	5	6	0.0
Speech Commands, 0.25	EcoFed	37.5 $\pm$ 1.3	42.8 $\pm$ 1.3	43.8 $\pm$ 1.5	11	12	2.1
	Oort-style	32.0 $\pm$ 1.1	40.1 $\pm$ 1.5	43.9 $\pm$ 1.1	19	20	6.0
	Energy-greedy	32.2 $\pm$ 1.4	39.6 $\pm$ 2.0	43.4 $\pm$ 0.9	19	20	6.0
	Static-Depth	35.1 $\pm$ 0.6	44.7 $\pm$ 2.3	46.4 $\pm$ 1.3	23	23	26.6
	FedAvg-Paced	33.1 $\pm$ 1.3	49.7 $\pm$ 2.4	51.7 $\pm$ 2.0	19	20	6.0
	FedAvg-Q8	32.7 $\pm$ 1.1	43.4 $\pm$ 2.6	47.0 $\pm$ 1.5	19	21	7.0
	FedProx	31.9 $\pm$ 1.1	40.6 $\pm$ 2.1	42.2 $\pm$ 1.9	19	20	6.0
	FedAvg	32.0 $\pm$ 1.3	40.0 $\pm$ 1.8	41.8 $\pm$ 2.0	19	20	6.0
Speech Commands, 1.0	EcoFed	46.6 $\pm$ 0.9	56.4 $\pm$ 1.7	57.8 $\pm$ 1.8	50	51	2.3
	Oort-style	46.1 $\pm$ 1.3	59.2 $\pm$ 2.9	63.7 $\pm$ 1.1	88	95	27.3
	Energy-greedy	47.4 $\pm$ 0.9	59.9 $\pm$ 1.5	62.5 $\pm$ 1.5	88	95	27.3
	Static-Depth	42.9 $\pm$ 1.4	59.3 $\pm$ 1.5	60.3 $\pm$ 1.5	91	95	34.4
	FedAvg-Paced	44.7 $\pm$ 1.0	61.0 $\pm$ 0.7	62.0 $\pm$ 1.2	90	95	27.2
	FedAvg-Q8	44.7 $\pm$ 1.0	61.0 $\pm$ 0.7	62.0 $\pm$ 1.2	90	95	27.2

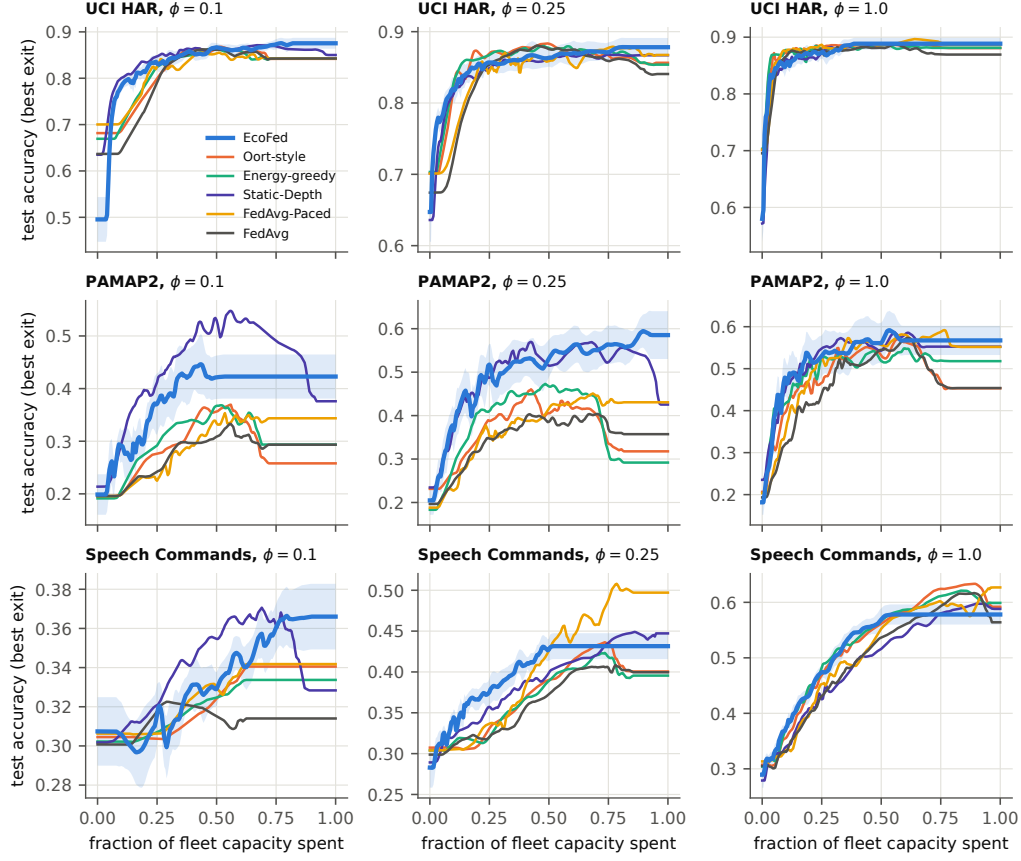


Figure 1: Test accuracy (best exit) against the fraction of fleet capacity spent; mean over 16 seeds, shaded band is the 95% CI of EcoFed. Methods that stop early keep their final accuracy. On the label-skewed PAMAP2 the energy-blind methods peak and then lose accuracy as batteries run out.

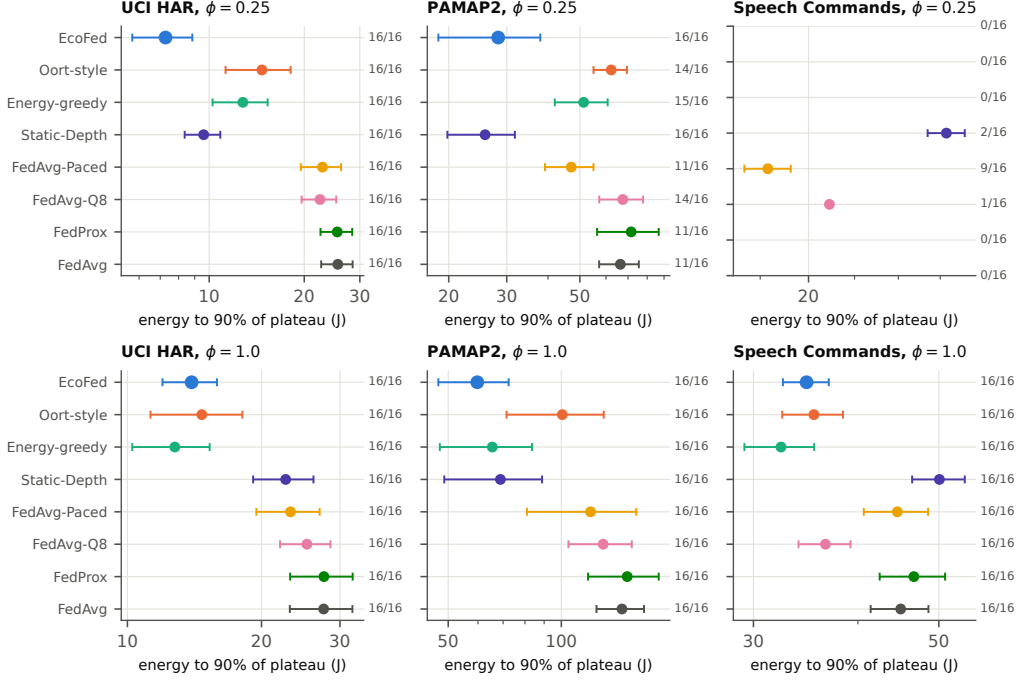


Figure 2: Fleet energy to reach 90% of the FedAvg plateau (mean with 95% CI over the seeds that reach it; the fraction on the right is the number of seeds that reach it). Speech Commands at $\phi = 0.1$ is omitted because no method reaches the target.

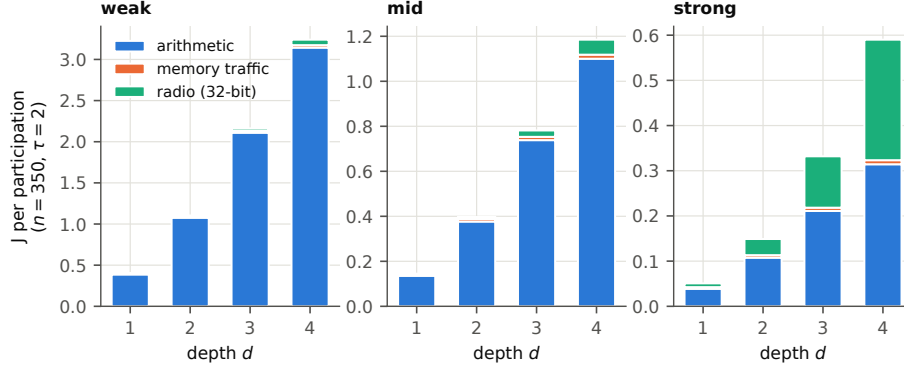


Figure 3: Modelled energy of one participation ($n = 350$ samples, 2 epochs, 32-bit upload) by depth, split into arithmetic, memory traffic and radio, for the three device classes. For this small network on-device arithmetic dominates; radio energy is visible only for the LTE-class device.

Table 5: Ablations of EcoFed (mean $\pm$ 95% CI over 8 seeds). Exit-4 acc.: test accuracy of the last exit, which is only trained by clients that reach full depth.

Data / ϕ	Variant	Acc@25%	Acc@100%	AUC	Exit-4 acc.	mean depth
UCI HAR, 0.1	EcoFed	83.4 $\pm$ 1.3	87.1 $\pm$ 0.9	83.1 $\pm$ 0.7	86.4 $\pm$ 0.9	1.80
	EcoFed w/o energy queue	55.2 $\pm$ 2.7	79.5 $\pm$ 3.3	70.0 $\pm$ 1.8	79.6 $\pm$ 3.8	1.59
	EcoFed w/o coverage queue	84.6 $\pm$ 1.5	85.0 $\pm$ 2.3	82.8 $\pm$ 1.8	76.4 $\pm$ 14.6	1.17
	EcoFed, fixed depth	76.2 $\pm$ 4.9	77.7 $\pm$ 3.5	73.7 $\pm$ 3.3	78.4 $\pm$ 3.0	1.01
	EcoFed, fixed precision	83.5 $\pm$ 2.3	86.8 $\pm$ 1.5	83.2 $\pm$ 1.3	86.3 $\pm$ 1.7	1.59
	EcoFed, fixed epochs	83.9 $\pm$ 1.8	86.2 $\pm$ 1.9	82.2 $\pm$ 1.2	85.2 $\pm$ 2.3	1.49
UCI HAR, 0.25	EcoFed	85.2 $\pm$ 1.1	87.4 $\pm$ 1.2	85.6 $\pm$ 1.1	87.7 $\pm$ 1.0	2.56
	EcoFed w/o energy queue	67.4 $\pm$ 4.6	80.8 $\pm$ 3.7	78.7 $\pm$ 1.7	81.2 $\pm$ 3.2	2.66
	EcoFed w/o coverage queue	84.8 $\pm$ 2.4	87.2 $\pm$ 2.0	86.3 $\pm$ 1.8	58.8 $\pm$ 27.1	1.23
	EcoFed, fixed depth	69.3 $\pm$ 4.1	69.3 $\pm$ 4.1	68.9 $\pm$ 4.0	61.6 $\pm$ 7.1	0.53
	EcoFed, fixed precision	86.0 $\pm$ 1.6	88.3 $\pm$ 1.3	85.6 $\pm$ 1.2	88.5 $\pm$ 1.4	2.37
	EcoFed, fixed epochs	84.9 $\pm$ 1.3	88.1 $\pm$ 1.5	85.4 $\pm$ 1.3	88.1 $\pm$ 1.7	2.33
PAMAP2, 0.1	EcoFed	36.7 $\pm$ 4.1	42.0 $\pm$ 4.5	38.5 $\pm$ 3.4	35.4 $\pm$ 5.3	1.76
	EcoFed w/o energy queue	25.9 $\pm$ 2.5	32.4 $\pm$ 13.3	33.6 $\pm$ 3.3	21.8 $\pm$ 7.5	1.40
	EcoFed w/o coverage queue	28.2 $\pm$ 6.9	27.5 $\pm$ 6.2	27.2 $\pm$ 5.9	14.0 $\pm$ 4.6	0.80
	EcoFed, fixed depth	26.9 $\pm$ 4.2	26.9 $\pm$ 4.2	26.4 $\pm$ 4.1	23.9 $\pm$ 4.3	0.55
	EcoFed, fixed precision	32.6 $\pm$ 5.5	38.9 $\pm$ 8.6	35.1 $\pm$ 7.2	35.3 $\pm$ 7.3	1.66
	EcoFed, fixed epochs	33.1 $\pm$ 6.3	33.8 $\pm$ 4.9	30.3 $\pm$ 5.0	27.0 $\pm$ 8.5	1.17
PAMAP2, 0.25	EcoFed	50.8 $\pm$ 4.8	57.5 $\pm$ 3.6	50.5 $\pm$ 3.7	49.2 $\pm$ 3.9	2.59
	EcoFed w/o energy queue	30.8 $\pm$ 3.3	28.9 $\pm$ 6.7	38.7 $\pm$ 4.6	23.7 $\pm$ 5.9	2.55
	EcoFed w/o coverage queue	50.1 $\pm$ 9.9	56.7 $\pm$ 8.6	52.3 $\pm$ 7.3	25.7 $\pm$ 3.7	1.64
	EcoFed, fixed depth	35.8 $\pm$ 4.1	40.9 $\pm$ 9.8	37.4 $\pm$ 7.2	32.7 $\pm$ 8.7	2.36
	EcoFed, fixed precision	47.8 $\pm$ 7.5	55.3 $\pm$ 6.4	49.4 $\pm$ 5.4	48.5 $\pm$ 4.8	2.54
	EcoFed, fixed epochs	40.5 $\pm$ 7.3	53.6 $\pm$ 7.5	45.4 $\pm$ 6.0	40.9 $\pm$ 6.6	2.51
Speech Commands, 0.1	EcoFed	31.0 $\pm$ 1.2	36.1 $\pm$ 1.2	33.4 $\pm$ 0.8	31.9 $\pm$ 3.4	1.04
	EcoFed w/o energy queue	28.7 $\pm$ 2.4	33.0 $\pm$ 0.8	30.6 $\pm$ 0.8	30.7 $\pm$ 2.9	0.78
	EcoFed w/o coverage queue	30.2 $\pm$ 2.9	39.8 $\pm$ 4.0	34.6 $\pm$ 2.0	30.6 $\pm$ 7.8	1.02
	EcoFed, fixed depth	29.7 $\pm$ 2.4	30.7 $\pm$ 3.5	30.1 $\pm$ 1.9	29.9 $\pm$ 4.2	0.83
	EcoFed, fixed precision	29.2 $\pm$ 2.3	34.6 $\pm$ 2.5	32.5 $\pm$ 1.1	30.8 $\pm$ 5.3	1.04
	EcoFed, fixed epochs	29.8 $\pm$ 1.9	35.3 $\pm$ 1.9	32.7 $\pm$ 1.2	32.0 $\pm$ 4.5	0.94
Speech Commands, 0.25	EcoFed	37.5 $\pm$ 1.3	42.8 $\pm$ 1.3	40.2 $\pm$ 1.1	41.0 $\pm$ 2.2	1.54
	EcoFed w/o energy queue	29.9 $\pm$ 2.1	34.6 $\pm$ 1.1	32.0 $\pm$ 0.9	33.2 $\pm$ 3.1	1.75
	EcoFed w/o coverage queue	39.8 $\pm$ 1.9	42.4 $\pm$ 3.0	40.4 $\pm$ 2.4	10.4 $\pm$ 8.2	1.01
	EcoFed, fixed depth	26.4 $\pm$ 6.4	26.4 $\pm$ 6.4	26.4 $\pm$ 6.4	21.7 $\pm$ 10.1	0.21
	EcoFed, fixed precision	37.3 $\pm$ 1.2	41.0 $\pm$ 1.8	38.5 $\pm$ 1.6	36.7 $\pm$ 3.6	1.26
	EcoFed, fixed epochs	39.4 $\pm$ 1.8	41.1 $\pm$ 1.6	39.6 $\pm$ 1.6	40.2 $\pm$ 2.5	1.26

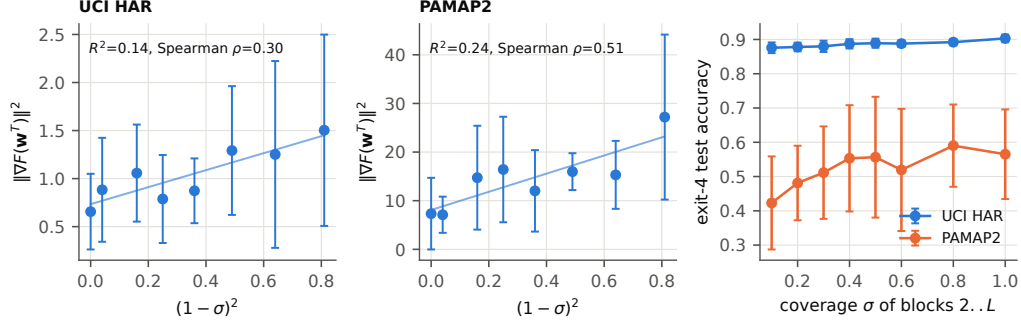


Figure 4: Coverage probe. Left and middle: squared full-objective gradient norm at the final iterate against $(1 - \sigma)^2$ (mean and 95% CI over 6 seeds, with a least-squares line over the individual runs). Right: last-exit test accuracy against the coverage σ of the deep blocks.

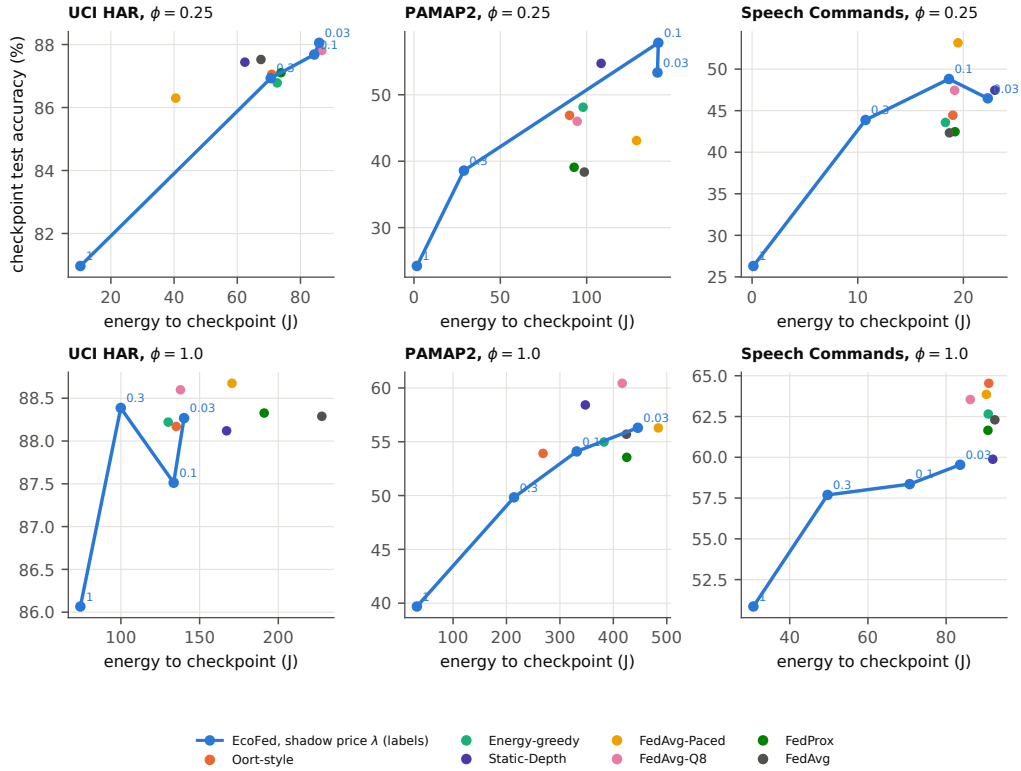


Figure 5: Energy-accuracy frontier at the validation-selected checkpoint. Line: EcoFed with shadow price $\lambda \in \{0.03, 0.1, 0.3, 1\}$ (labels); dots: baselines at their tuned settings. All points are means over the first eight seeds.

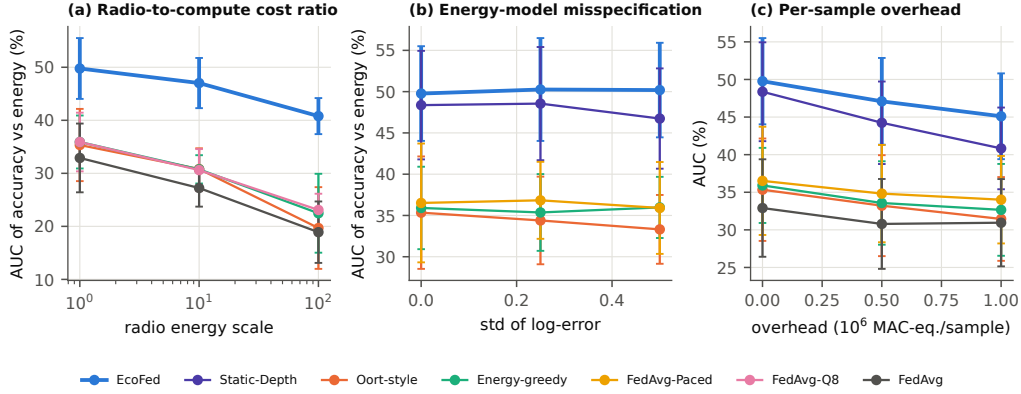


Figure 6: Sensitivity of the area under the accuracy-versus-energy curve on PAMAP2 at $\phi = 0.25$ (mean and 95% CI over 8 seeds). (a) Radio energy scaled by 1, 10, 100. (b) Log-normal error between the energy constants assumed by the controller and the true ones. (c) Unmodelled per-sample overhead added to the true energy.

Table 6: Lifecycle energy (means over 16 seeds). Test acc.: accuracy of the deployed cascade (single exit for the first row). E_{train} : fleet energy used for training. $\bar{e}_{inf}$: expected inference energy per window. $M_{\times} = E_{train}/\bar{e}_{inf}$: number of inferences after which serving costs as much as training did.

Data / ϕ	Model	Test acc. (%)	E_{train} (J)	$\bar{e}_{inf}$ (μ J)	exits at 1 (%)	$M_{\times}$ (10^5)	days to $M_{\times}$
UCI HAR, 0.25	FedAvg, single exit	87.6	116.1	845	0	1.37	0.13
	FedAvg, cascade	83.2	116.1	270	23	4.30	0.40
	EcoFed, cascade	86.9	101.3	118	96	8.60	0.80
UCI HAR, 1.0	FedAvg, single exit	89.2	378.9	845	0	4.49	0.42
	FedAvg, cascade	86.0	378.9	215	44	17.61	1.63
	EcoFed, cascade	88.0	186.5	130	94	14.38	1.33
PAMAP2, 0.25	FedAvg, single exit	22.1	136.2	950	0	1.43	0.11
	FedAvg, cascade	32.5	136.2	422	64	3.22	0.24
	EcoFed, cascade	58.5	163.4	219	97	7.47	0.55
PAMAP2, 1.0	FedAvg, single exit	39.9	555.9	950	0	5.85	0.43
	FedAvg, cascade	45.7	555.9	399	70	13.94	1.03
	EcoFed, cascade	57.9	414.5	382	70	10.85	0.80
Speech Commands, 0.25	FedAvg, single exit	35.2	19.8	454	0	0.44	0.01
	FedAvg, cascade	39.4	19.8	386	0	0.51	0.01
	EcoFed, cascade	42.4	12.0	331	41	0.36	0.01
Speech Commands, 1.0	FedAvg, single exit	57.6	95.0	454	0	2.09	0.05
	FedAvg, cascade	55.3	95.0	368	0	2.58	0.06
	EcoFed, cascade	56.9	51.0	349	17	1.46	0.04

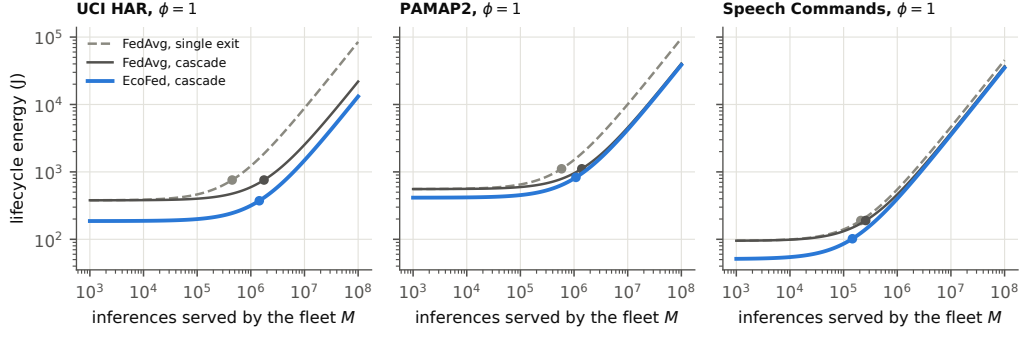


Figure 7: Lifecycle energy against the number of inferences served by the fleet ($\phi = 1$). Dots mark M_x , where inference energy equals training energy.

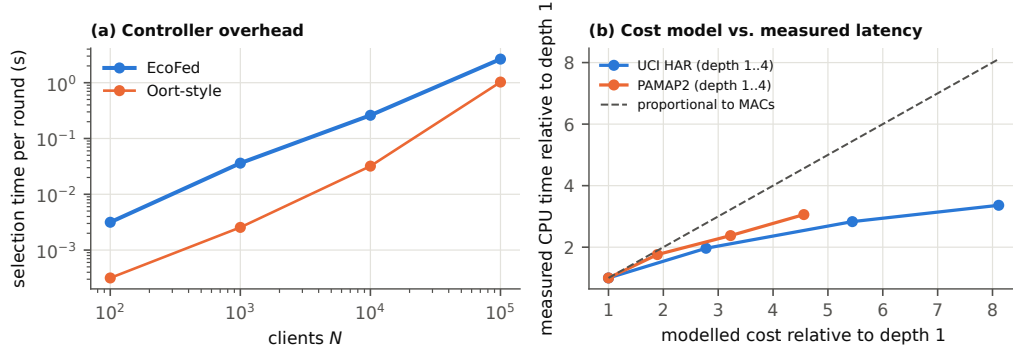


Figure 8: (a) Time of the selection step against the number of clients (5% selected; synthetic fleets; one CPU core). (b) Measured CPU time per training sample against the modelled cost, both relative to depth 1; the dashed line is proportionality.

Table B.7: Selected hyper-parameters per method, dataset and budget from 14 random-search trials each (mean validation AUC over the two tuning seeds); lr: learning rate. μ : FedProx weight; ϵ , ξ : exploration fraction and speed exponent; EcoFed: $\text{rho_min} = \rho^{\min}$, $\text{q_floor} = \lambda$, $\text{gamma_pow} = \kappa_d$, $V = 2$.

Data / ϕ	Method	Knobs	lr	Validation AUC (%)
UCI HAR, 0.1	EcoFed	V=2.0, gamma_pow=0.5, q_floor=0.1, rho_min=0.3	0.626	87.1
	Oort-style	eps=0.1, xi=2.0	0.112	86.0
	Energy-greedy	eps=0.1	0.143	86.5
	Static-Depth	–	0.8	88.6
	FedAvg-Paced	–	0.087	85.3
	FedAvg-Q8	–	0.143	84.4
	FedProx	mu=0.001	0.087	83.6
	FedAvg	–	0.143	83.0
UCI HAR, 0.25	EcoFed	V=2.0, gamma_pow=0.25, q_floor=0.3, rho_min=0.6	0.299	88.9
	Oort-style	eps=0.1, xi=2.0	0.087	87.9
	Energy-greedy	eps=0.1	0.112	88.8
	Static-Depth	–	0.489	89.8
	FedAvg-Paced	–	0.068	86.8
	FedAvg-Q8	–	0.068	86.5
	FedProx	mu=0.001	0.042	86.1
	FedAvg	–	0.042	86.2
PAMAP2, 0.1	EcoFed	V=2.0, gamma_pow=0.5, q_floor=0.1, rho_min=0.6	0.143	46.7
	Oort-style	eps=0.1, xi=2.0	0.068	38.7
	Energy-greedy	eps=0.3	0.143	38.7
	Static-Depth	–	0.8	52.2
	FedAvg-Paced	–	0.489	36.4
	FedAvg-Q8	–	0.087	37.8
	FedProx	mu=0.01	0.068	37.3
	FedAvg	–	0.489	37.8
PAMAP2, 0.25	EcoFed	V=2.0, gamma_pow=0.25, q_floor=0.1, rho_min=0.6	0.489	57.6
	Oort-style	eps=0.3, xi=1.0	0.112	44.7
	Energy-greedy	eps=0.1	0.087	45.0
	Static-Depth	–	0.8	53.8
	FedAvg-Paced	–	0.489	46.9
	FedAvg-Q8	–	0.234	45.0
	FedProx	mu=0.01	0.626	43.6
	FedAvg	–	0.8	44.3
Speech Commands, 0.1	EcoFed	V=2.0, gamma_pow=0.5, q_floor=0.03, rho_min=0.6	0.626	27.7
	Oort-style	eps=0.1, xi=2.0	0.068	25.8
	Energy-greedy	eps=0.1	0.143	25.7
	Static-Depth	–	0.383	27.8
	FedAvg-Paced	–	0.053	25.7
	FedAvg-Q8	–	0.112	27.4
	FedProx	mu=0.1	0.112	26.0
	FedAvg	–	0.068	26.2
Speech Commands, 0.25	EcoFed	V=2.0, gamma_pow=0.25, q_floor=0.3, rho_min=0.3	0.8	37.2
	Oort-style	eps=0.1, xi=1.0	0.087	30.0
	Energy-greedy	eps=0.1	0.068	31.0
	Static-Depth	–	0.299	34.0
	FedAvg-Paced	–	0.042	34.7
	FedAvg-Q8	–	0.087	31.8
	FedProx	mu=0.1	0.087	29.5
	FedAvg	–	0.112	29.9

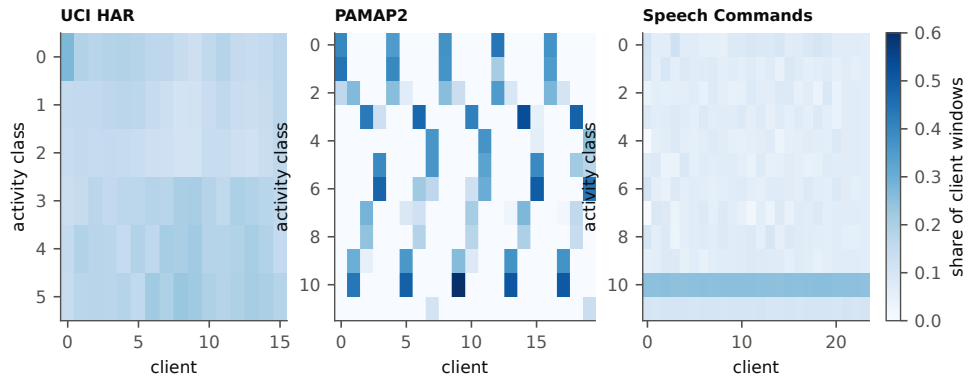


Figure B.9: Share of each activity class in each client’s training windows. UCI HAR clients are persons who all performed all activities; PAMAP2 clients are time chunks of a sequential protocol and hold three to five classes; Speech Commands clients are speakers (class 10 is the capped “unknown” class, class 11 silence).

Table B.8: Best-checkpoint accuracy, EcoFed minus baseline (pp).

Data / ϕ	vs.	mean Δ (pp)	95% CI	n	Holm p
UCI HAR, 0.1	Oort-style	0.28	[-1.39, 2.01]	16	1.000
	Energy-greedy	-0.51	[-2.24, 1.30]	16	1.000
	Static-Depth	0.81	[-0.77, 2.41]	16	1.000
	FedAvg-Paced	-0.19	[-1.53, 1.31]	16	1.000
	FedAvg-Q8	0.31	[-1.25, 1.74]	16	1.000
	FedProx	-0.22	[-1.60, 1.18]	16	1.000
	FedAvg	-1.11	[-2.35, -0.04]	16	0.639
UCI HAR, 0.25	Oort-style	0.10	[-1.25, 1.42]	16	1.000
	Energy-greedy	0.01	[-1.37, 1.39]	16	1.000
	Static-Depth	0.29	[-1.19, 1.64]	16	1.000
	FedAvg-Paced	0.44	[-0.95, 1.77]	16	1.000
	FedAvg-Q8	-0.39	[-1.93, 1.03]	16	1.000
	FedProx	0.20	[-1.01, 1.25]	16	1.000
	FedAvg	-0.02	[-1.49, 1.31]	16	1.000
UCI HAR, 1.0	Oort-style	-0.28	[-1.56, 0.93]	16	1.000
	Energy-greedy	-0.35	[-1.62, 0.90]	16	1.000
	Static-Depth	-0.17	[-1.39, 1.16]	16	1.000
	FedAvg-Paced	-0.96	[-2.00, 0.08]	16	0.721
	FedAvg-Q8	-0.60	[-1.78, 0.51]	16	1.000
	FedProx	-0.33	[-1.48, 0.87]	16	1.000
	FedAvg	-0.22	[-1.49, 1.02]	16	1.000
PAMAP2, 0.1	Oort-style	4.41	[-1.41, 11.07]	16	0.410
	Energy-greedy	2.64	[-2.89, 8.63]	16	0.410
	Static-Depth	-12.48*	[-17.94, -6.98]	16	0.004
	FedAvg-Paced	5.26	[0.91, 9.91]	16	0.254
	FedAvg-Q8	6.85	[0.32, 13.97]	16	0.351
	FedProx	5.05	[-0.46, 11.64]	16	0.410
	FedAvg	5.37	[-0.00, 11.60]	16	0.399
PAMAP2, 0.25	Oort-style	10.58*	[5.44, 15.40]	16	0.003
	Energy-greedy	9.30*	[5.24, 12.92]	16	0.001
	Static-Depth	1.13	[-3.28, 5.34]	16	0.627
	FedAvg-Paced	12.01*	[7.52, 16.90]	16	0.001
	FedAvg-Q8	12.76*	[7.99, 17.66]	16	0.001
	FedProx	16.09*	[12.24, 20.64]	16	0.000
	FedAvg	16.95*	[11.29, 22.34]	16	0.000
PAMAP2, 1.0	Oort-style	1.32	[-2.99, 5.63]	16	1.000
	Energy-greedy	2.43	[-2.16, 7.20]	16	1.000
	Static-Depth	-3.98	[-8.49, 0.64]	16	0.740
	FedAvg-Paced	-2.48	[-6.20, 1.18]	16	1.000
	FedAvg-Q8	-5.02	[-8.88, -1.36]	16	0.158
	FedProx	-0.05	[-3.59, 3.72]	16	1.000
	FedAvg	-2.58	[-7.41, 1.98]	16	1.000
Speech Commands, 0.1	Oort-style	2.19	[-0.11, 4.22]	16	0.346
	Energy-greedy	3.38	[0.87, 5.75]	16	0.115
	Static-Depth	-0.83	[-3.46, 2.36]	16	0.603
	FedAvg-Paced	2.59	[-0.62, 5.87]	16	0.479
	FedAvg-Q8	1.83	[-0.49, 4.53]	16	0.479
	FedProx	2.79	[0.21, 5.54]	16	0.346
	FedAvg	5.89*	[2.41, 9.21]	16	0.037
Speech Commands, 0.25	Oort-style	-0.07	[-1.60, 1.48]	16	1.000
	Energy-greedy	0.39	[-0.96, 1.77]	16	1.000
	Static-Depth	-2.58*	[-3.99, -1.14]	16	0.020
	FedAvg-Paced	-7.83*	[-9.83, -5.90]	16	0.000

Table B.9: Final accuracy, EcoFed minus baseline (pp).

Data / ϕ	vs.	mean Δ (pp)	95% CI	n	Holm p
UCI HAR, 0.1	Oort-style	2.89*	[1.28, 4.42]	16	0.017
	Energy-greedy	2.84*	[1.21, 4.48]	16	0.017
	Static-Depth	2.06*	[0.93, 3.20]	16	0.017
	FedAvg-Paced	2.84*	[1.46, 4.37]	16	0.012
	FedAvg-Q8	3.20*	[1.84, 4.72]	16	0.005
	FedProx	1.84*	[0.56, 3.19]	16	0.037
	FedAvg	2.82*	[0.85, 4.95]	16	0.037
UCI HAR, 0.25	Oort-style	1.73	[0.38, 3.09]	16	0.083
	Energy-greedy	1.99	[0.63, 3.39]	16	0.060
	Static-Depth	0.54	[-0.53, 1.55]	16	0.563
	FedAvg-Paced	0.71	[-0.52, 1.89]	16	0.563
	FedAvg-Q8	2.51*	[1.83, 3.27]	16	0.000
	FedProx	2.87*	[1.71, 4.15]	16	0.002
	FedAvg	3.30*	[2.16, 4.54]	16	0.001
UCI HAR, 1.0	Oort-style	0.07	[-0.99, 1.03]	16	1.000
	Energy-greedy	0.11	[-0.73, 1.01]	16	1.000
	Static-Depth	-0.25	[-1.27, 0.78]	16	1.000
	FedAvg-Paced	-0.67	[-1.59, 0.18]	16	0.710
	FedAvg-Q8	0.94	[0.10, 1.80]	16	0.280
	FedProx	1.37	[0.50, 2.26]	16	0.069
	FedAvg	1.37	[0.52, 2.28]	16	0.069
PAMAP2, 0.1	Oort-style	16.24*	[12.11, 20.44]	16	0.000
	Energy-greedy	12.67*	[7.74, 17.42]	16	0.001
	Static-Depth	4.44*	[2.07, 6.89]	16	0.004
	FedAvg-Paced	7.79*	[4.83, 10.87]	16	0.001
	FedAvg-Q8	12.49*	[8.44, 16.10]	16	0.000
	FedProx	13.50*	[10.02, 16.76]	16	0.000
	FedAvg	12.69*	[9.02, 16.25]	16	0.000
PAMAP2, 0.25	Oort-style	25.76*	[21.38, 30.35]	16	0.000
	Energy-greedy	28.34*	[24.11, 33.26]	16	0.000
	Static-Depth	14.99*	[11.61, 18.25]	16	0.000
	FedAvg-Paced	14.69*	[10.32, 19.38]	16	0.000
	FedAvg-Q8	22.79*	[18.99, 26.46]	16	0.000
	FedProx	21.67*	[17.62, 25.56]	16	0.000
	FedAvg	21.80*	[17.87, 25.59]	16	0.000
PAMAP2, 1.0	Oort-style	11.16*	[6.75, 15.95]	16	0.002
	Energy-greedy	5.73*	[2.02, 9.17]	16	0.026
	Static-Depth	-0.14	[-3.10, 2.76]	16	1.000
	FedAvg-Paced	0.02	[-2.59, 2.61]	16	1.000
	FedAvg-Q8	8.61*	[4.00, 14.22]	16	0.024
	FedProx	9.62*	[5.58, 13.76]	16	0.002
	FedAvg	11.80*	[7.14, 16.78]	16	0.002
Speech Commands, 0.1	Oort-style	2.07	[0.72, 3.43]	16	0.058
	Energy-greedy	2.76	[0.91, 4.65]	16	0.058
	Static-Depth	3.29	[0.46, 6.44]	16	0.114
	FedAvg-Paced	1.95	[-0.47, 4.71]	16	0.184
	FedAvg-Q8	1.63	[0.62, 2.73]	16	0.058
	FedProx	1.98	[0.58, 3.50]	16	0.071
	FedAvg	4.72*	[2.31, 7.13]	16	0.014
Speech Commands, 0.25	Oort-style	2.73*	[1.12, 4.38]	16	0.020
	Energy-greedy	3.24*	[1.44, 4.96]	16	0.020
	Static-Depth	-1.93	[-3.76, -0.09]	16	0.159
	FedAvg-Paced	-6.92*	[-8.88, -4.90]	16	0.000

Table B.10: Total fleet energy used, ratio baseline/EcoFed.

Data / ϕ	vs.	mean ratio (baseline/EcoFed)	95% CI	n	Holm p
UCI HAR, 0.1	Oort-style	0.94*	[0.91, 0.96]	16	0.000
	Energy-greedy	0.94*	[0.91, 0.96]	16	0.000
	Static-Depth	1.25*	[1.22, 1.28]	16	0.000
	FedAvg-Paced	0.94*	[0.91, 0.96]	16	0.000
	FedAvg-Q8	0.95*	[0.93, 0.98]	16	0.002
	FedProx	0.94*	[0.91, 0.96]	16	0.000
	FedAvg	0.94*	[0.91, 0.96]	16	0.000
UCI HAR, 0.25	Oort-style	1.15*	[1.13, 1.16]	16	0.000
	Energy-greedy	1.15*	[1.13, 1.16]	16	0.000
	Static-Depth	1.06*	[1.04, 1.09]	16	0.000
	FedAvg-Paced	1.15*	[1.13, 1.16]	16	0.000
	FedAvg-Q8	1.18*	[1.17, 1.20]	16	0.000
	FedProx	1.15*	[1.13, 1.16]	16	0.000
	FedAvg	1.15*	[1.13, 1.16]	16	0.000
UCI HAR, 1.0	Oort-style	2.03*	[2.00, 2.05]	16	0.000
	Energy-greedy	1.89*	[1.87, 1.91]	16	0.000
	Static-Depth	1.45*	[1.43, 1.47]	16	0.000
	FedAvg-Paced	2.09*	[2.06, 2.12]	16	0.000
	FedAvg-Q8	1.89*	[1.87, 1.92]	16	0.000
	FedProx	2.03*	[2.01, 2.06]	16	0.000
	FedAvg	2.03*	[2.01, 2.06]	16	0.000
PAMAP2, 0.1	Oort-style	1.49*	[1.40, 1.58]	16	0.000
	Energy-greedy	1.49*	[1.40, 1.58]	16	0.000
	Static-Depth	2.04*	[1.96, 2.11]	16	0.000
	FedAvg-Paced	1.49*	[1.40, 1.58]	16	0.000
	FedAvg-Q8	1.53*	[1.43, 1.62]	16	0.000
	FedProx	1.49*	[1.40, 1.58]	16	0.000
	FedAvg	1.49*	[1.40, 1.58]	16	0.000
PAMAP2, 0.25	Oort-style	0.83*	[0.82, 0.85]	16	0.000
	Energy-greedy	0.83*	[0.82, 0.85]	16	0.000
	Static-Depth	1.05*	[1.04, 1.06]	16	0.000
	FedAvg-Paced	0.83*	[0.82, 0.85]	16	0.000
	FedAvg-Q8	0.83*	[0.82, 0.85]	16	0.000
	FedProx	0.83*	[0.82, 0.85]	16	0.000
	FedAvg	0.83*	[0.82, 0.85]	16	0.000
PAMAP2, 1.0	Oort-style	1.35*	[1.34, 1.37]	16	0.000
	Energy-greedy	1.25*	[1.23, 1.27]	16	0.000
	Static-Depth	1.13*	[1.11, 1.16]	16	0.000
	FedAvg-Paced	1.38*	[1.36, 1.39]	16	0.000
	FedAvg-Q8	1.27*	[1.26, 1.28]	16	0.000
	FedProx	1.34*	[1.33, 1.36]	16	0.000
	FedAvg	1.34*	[1.33, 1.36]	16	0.000
Speech Commands, 0.1	Oort-style	0.73*	[0.71, 0.75]	16	0.000
	Energy-greedy	0.73*	[0.71, 0.75]	16	0.000
	Static-Depth	1.04*	[1.01, 1.06]	16	0.018
	FedAvg-Paced	0.73*	[0.71, 0.75]	16	0.000
	FedAvg-Q8	0.79*	[0.77, 0.80]	16	0.000
	FedProx	0.73*	[0.71, 0.75]	16	0.000
	FedAvg	0.73*	[0.71, 0.75]	16	0.000
Speech Commands, 0.25	Oort-style	1.66*	[1.60, 1.72]	16	0.000
	Energy-greedy	1.66*	[1.60, 1.72]	16	0.000
	Static-Depth	1.95*	[1.90, 2.02]	16	0.000
	FedAvg-Paced	1.66*	[1.60, 1.72]	16	0.000
	FedAvg-Q8	1.74*	[1.69, 1.81]	16	0.000
	FedProx	1.66*	[1.60, 1.72]	16	0.000
	FedAvg	1.66*	[1.60, 1.72]	16	0.000
Speech Commands, 1.0	Oort-style	1.87*	[1.83, 1.90]	16	0.000
	Energy-greedy	1.87*	[1.83, 1.90]	16	0.000
	Static-Depth	1.86*	[1.83, 1.90]	16	0.000
	FedAvg-Paced	1.87*	[1.83, 1.90]	16	0.000
	FedAvg-Q8	1.81*	[1.78, 1.83]	16	0.000
	FedProx	1.87*	[1.83, 1.90]	16	0.000
	FedAvg	1.87*	[1.83, 1.90]	16	0.000

Table B.11: Energy to reach 90% of the plateau, ratio baseline/EcoFed (seeds that reach it in both).

Data / ϕ	vs.	mean ratio (baseline/EcoFed)	95% CI	n	Holm p
UCI HAR, 0.1	Oort-style	2.54*	[2.00, 3.14]	16	0.000
	Energy-greedy	2.38*	[1.84, 3.10]	16	0.000
	Static-Depth	0.96	[0.78, 1.20]	16	0.219
	FedAvg-Paced	1.66*	[1.27, 2.13]	16	0.017
	FedAvg-Q8	2.67*	[2.11, 3.47]	16	0.000
	FedProx	2.26*	[1.68, 2.94]	16	0.002
	FedAvg	2.55*	[1.92, 3.27]	16	0.000
UCI HAR, 0.25	Oort-style	2.17*	[1.69, 2.73]	16	0.000
	Energy-greedy	2.24*	[1.60, 2.93]	16	0.020
	Static-Depth	1.48*	[1.23, 1.77]	16	0.006
	FedAvg-Paced	3.80*	[2.92, 4.75]	16	0.000
	FedAvg-Q8	3.58*	[2.83, 4.41]	16	0.000
	FedProx	4.22*	[3.24, 5.24]	16	0.000
	FedAvg	4.23*	[3.27, 5.21]	16	0.000
UCI HAR, 1.0	Oort-style	1.19	[0.84, 1.59]	16	0.906
	Energy-greedy	0.98	[0.77, 1.19]	16	0.626
	Static-Depth	1.71*	[1.45, 2.01]	16	0.000
	FedAvg-Paced	1.80*	[1.44, 2.17]	16	0.002
	FedAvg-Q8	2.00*	[1.59, 2.46]	16	0.000
	FedProx	2.13*	[1.68, 2.63]	16	0.000
	FedAvg	2.13*	[1.69, 2.64]	16	0.000
PAMAP2, 0.1	Oort-style	4.00	[1.84, 6.21]	6	0.110
	Energy-greedy	2.85	[1.91, 4.58]	5	0.110
	Static-Depth	1.79	[1.05, 2.83]	9	0.261
	FedAvg-Paced	3.53	[1.73, 6.60]	5	0.125
	FedAvg-Q8	3.71	[1.69, 6.00]	6	0.112
	FedProx	3.87	[1.86, 7.68]	3	0.261
	FedAvg	4.64	[2.16, 8.24]	5	0.112
PAMAP2, 0.25	Oort-style	2.90*	[2.49, 3.33]	14	0.000
	Energy-greedy	2.10*	[1.77, 2.48]	15	0.000
	Static-Depth	1.04	[0.82, 1.25]	16	0.556
	FedAvg-Paced	2.41*	[1.94, 2.85]	11	0.000
	FedAvg-Q8	3.16*	[2.29, 4.19]	14	0.000
	FedProx	3.75*	[2.86, 4.86]	11	0.000
	FedAvg	3.47*	[2.65, 4.36]	11	0.000
PAMAP2, 1.0	Oort-style	1.74*	[1.40, 2.08]	16	0.003
	Energy-greedy	1.19	[0.91, 1.48]	16	0.675
	Static-Depth	1.15	[0.99, 1.32]	16	0.527
	FedAvg-Paced	2.09*	[1.71, 2.54]	16	0.000
	FedAvg-Q8	2.30*	[1.98, 2.66]	16	0.000
	FedProx	2.58*	[2.25, 2.97]	16	0.000
	FedAvg	2.60*	[2.28, 2.95]	16	0.000
Speech Commands, 0.1	Oort-style	nan	—	0	1.000
	Energy-greedy	nan	—	0	1.000
	Static-Depth	nan	—	0	1.000
	FedAvg-Paced	nan	—	0	1.000
	FedAvg-Q8	nan	—	0	1.000
	FedProx	nan	—	0	1.000
	FedAvg	nan	—	0	1.000
Speech Commands, 0.25	Oort-style	nan	—	0	1.000
	Energy-greedy	nan	—	0	1.000
	Static-Depth	nan	—	0	1.000
	FedAvg-Paced	nan	—	0	1.000
	FedAvg-Q8	nan	—	0	1.000
	FedProx	nan	—	0	1.000
	FedAvg	nan	—	0	1.000
Speech Commands, 1.0	Oort-style	1.03	[0.95, 1.11]	16	0.697
	Energy-greedy	0.94	[0.86, 1.02]	16	0.317
	Static-Depth	1.46*	[1.34, 1.60]	16	0.000
	FedAvg-Paced	1.29*	[1.19, 1.40]	16	0.000
	FedAvg-Q8	1.27*	[1.17, 1.37]	16	0.000

Table B.12: Test accuracy (%) of each exit at the end of training, $\phi = 0.25$, mean over 16 seeds. Single-exit FedAvg trains only the last exit.

Data / ϕ	Method	Exit 1	Exit 2	Exit 3	Exit 4
UCI HAR, 0.25	EcoFed	86.1	86.5	87.4	87.7
	Static-Depth	86.8	85.1	85.6	80.7
	Oort-style	61.8	83.5	84.9	85.6
	FedAvg	51.5	82.3	83.9	84.5
	FedAvg (single exit)	16.1	16.0	14.5	87.6
PAMAP2, 0.25	EcoFed	57.9	54.6	52.2	49.2
	Static-Depth	44.9	34.5	30.6	28.6
	Oort-style	26.1	32.8	30.6	30.1
	FedAvg	35.7	32.2	28.7	27.1
	FedAvg (single exit)	9.1	9.0	5.4	22.1
Speech Commands, 0.25	EcoFed	38.7	39.8	40.7	41.0
	Static-Depth	40.2	44.7	7.8	9.9
	Oort-style	14.1	33.8	37.1	40.0
	FedAvg	17.1	33.2	38.0	39.5
	FedAvg (single exit)	6.9	8.9	7.3	35.2